Monetary and Exchange Rate Policies in a Global Economy*

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Abstract

A consensus in the small open economy literature is that optimal monetary policy and foreign exchange intervention (FXI) separately stabilize inflation and the exchange rate. I develop an analytically tractable two-country framework where FXI balances internal and external objectives. Under international policy cooperation, optimal FXI mitigates the trade-off between domestic inflation and demand faced by monetary authorities, allowing them to stabilize inflation with moderate changes in the interest rate. At the same time, optimal FXI targets world demand since it affects international prices. The model thus suggests a novel complementarity between conventional monetary policy and unconventional exchange rate policy tools and provides a rationale for their combined use. I further show that FXI contributes more to domestic inflation stabilization when all goods traded in international markets are priced in dollars.

Keywords: Capital flows, International risk-sharing, Foreign exchange intervention, Optimal targeting rules, International policy cooperation.

JEL Classification Codes: E58, F31, F32, F41, F42.

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1 Introduction

The classical "trilemma" (Mundell 1957, Fleming 1962) suggests that, under free capital mobility, monetary authorities cannot simultaneously pursue inflation/output and exchange rate stabilization. In a modern, financially globalized world, countries increasingly rely on unconventional policy tools to manage the capital account and insulate themselves from international spillovers of shocks and crises to preserve their monetary autonomy (Rey 2015, Kalemli-Özcan 2019, Miranda-Agrippino and Rey 2020). In particular, many countries resort to foreign exchange intervention (FXI), i.e. purchases or sales of foreign currency reserves by central banks, with over 100 countries regularly using FXI (Adler et al. 2023).[1]

There have been recent discussions on the optimal mix of monetary and exchange rate policies. In the small-open economy literature, monetary policy and FXI are often considered to be two distinct policies and operate separately on their objectives: FXI exclusively stabilizes the exchange rate while monetary policy independently stabilizes inflation (Cavallino 2019, Basu et al. 2020, Itskhoki and Mukhin 2023). This rationale is based on a small-open economy case that takes the international prices of goods as given.

This paper studies two features of FXI that became salient during the COVID-19 pandemic and the Russian-Ukrainian war. These features depart from the consensus in the literature. First, central banks use FXI to stabilize inflation. For example, the pandemic and the war caused worldwide inflationary pressure due to supply chain disruption. However, China and India set a record low of around 4% interest rate during the pandemic. Brazil also lowered its rate to 2% and Japan even adopted a negative interest rate policy. Instead, they responded to the crisis by selling dollar reserves: China and Brazil sold 38 billion and 25 billion dollars in 2020, and India and Japan sold 32 billion and 63 billion dollars in 2022, respectively. This approach contrasts with the conventional method of combating inflation by raising the policy rate. This example suggests that the objectives of monetary policy and FXI are not necessarily independent from each other.[2]

Second, large open economies intervene in the FX market on massive scales. Most notably,

[1]In countries such as Japan, the finance ministry is in charge of FXI instead of the central bank. This paper considers a joint government consisting of the central bank and the finance ministry.

[2]The empirical analysis on the macroeconomic effects of FXI requires a rigorous identification of FXI. As discussed in Fratzscher et al. (2019) and Maggiori (2022), this is a challenging task in the literature and beyond the scope of this paper. As a step toward identification, Gonzalez et al. (2021) and Rodnyansky et al. (2024) combine FXI data with granular bank- or firm-level balance sheet information and study within-country variation of responses to FXI.

China has 3.5 trillion dollars and Japan has 1.4 trillion dollars of FX reserves, accounting for nearly 40% of the world's FX reserves (IMF International Financial Statistics). These countries are on the US monitoring list for accumulating excess FX reserves and gaining an unfair competitive advantage in international trade (US Department of the Treasury 2024). Moreover, large countries, such as China, India, and Japan, sold FX reserves to protect their currency during the pandemic and the war, as noted above. In such a large open-economy context, exchange rate manipulation may lead to global economic fluctuation by changing the international price and the world demand for domestic and foreign goods. This paper studies whether policies should respond to domestic inflation and output gap or global business cycle conditions and imbalances beyond domestic objectives.

This paper constructs a New Keynesian model that studies the optimality, interactions, and trade-offs of monetary and exchange rate policies. The model features two large countries: the US and the local; two policies: monetary policy and FXI; and two frictions: nominal and financial frictions. The frictions I introduce imply a non-neutral role for monetary policy and FXI: sticky prices allow monetary policy to influence the real interest rate (Calvo 1983) and limits to arbitrage in international capital markets allow FXI to influence the exchange rate by changing the demand and supply of bonds in different currencies (Gabaix and Maggiori 2015, Itskhoki and Mukhin 2021).[3]

My first contribution is to provide a full analytical characterization of optimal monetary policy and FXI rules in a large two-country framework. I begin with an international cooperation case, in which central banks maximize global welfare.[4] The key finding is that FXI trades off an internal objective (domestic inflation and demand) and an external objective (foreign inflation and demand).

On the one hand, FXI helps central banks stabilize inflation. When a central bank purchases local currency bonds using FXI, the demand for the local currency increases, so the local currency appreciates. As a result, local goods become more expensive for US households, making them consume local goods less. This reduced US demand, in turn, leads to lower prices and lower inflation of local goods. Hence, FXI allows central banks to stabilize inflation

[3] Models of limits to arbitrage are motivated by empirical literature on forward premium puzzle (Fama 1984). Data shows that cross-currency interest differentials are not offset by expected exchange rate depreciation, resulting in positive excess returns on currency carry trades. Without limits to arbitrage, households would obtain an infinite carry-trade profit by investing in currencies with higher returns.

[4] In the later section, I contrast the optimal policies under cooperative and non-cooperative equilibria and study gains from cooperation.

even without tightening the monetary policy and depressing local consumption demand. This rationalizes why central banks used FXI rather than monetary policy to stabilize inflation during the pandemic and war.

On the other hand, FXI distorts world inflation and demand. To understand the concept, consider an increase in local output, which increases local consumption. At the same time, a higher supply depreciates the local exchange rate and enables US households to import local goods at lower prices. Hence, US consumption also increases. Exchange rates adjust automatically and smooth consumption across countries, even if households cannot trade state-contingent assets internationally.

However, FXI distorts this consumption smoothing by manipulating the exchange rate. When the central bank purchases the local currency, the local currency appreciates. This appreciation has asymmetric effects on local and US households: local households import US goods at lower prices and increase consumption, while US households import local goods at higher prices and decrease consumption. As a result, FXI disproportionately benefits the local households over the US households.

My analytical result shows that FXI has two main objectives. First, FXI stabilizes non-fundamental volatility in capital flows and exchange rates, as pointed out in the literature of small-open economy (Cavallino 2019, Basu et al. 2020, Itskhoki and Mukhin 2023). Second, when local inflation is higher than US inflation, the optimal policy is to buy the local currency, which stabilizes domestic inflation. However, this comes at the cost of destabilizing world demand since FXI by large-open economies affects international prices. Hence, optimal FXI under cooperation balances internal objectives (domestic inflation and consumption demand) and external objectives (relative demand across countries).

I then calibrate the model and explore the quantitative implications of my model. Without FXI, external inflationary cost-push shocks weaken the independence of monetary policy, making it difficult to stabilize domestic inflation and output simultaneously. However, FXI improves the independence of monetary policy, allowing monetary authorities to achieve nearly full stabilization of inflation and output with little changes in the monetary policy rate. The key policy implication is that monetary policy and FXI are not completely independent and cannot be optimized in isolation, but a policy encompassing both instruments achieves the optimal outcome. In other words, monetary policy and FXI are complements.

Having highlighted the price stabilization channel of FXI, my second contribution is to establish a novel relationship between capital flow management in international finance and the US dollar's dominance in international trade. These two strands of literature in international macroeconomics are often discussed in separate contexts. My paper aims to bridge the gap between them. Recent evidence suggests that the majority of world trade is invoiced in a small number of dominant currencies, particularly the US dollar (Gopinath et al. 2020). Motivated by this fact, I consider a case where all goods traded in international markets are priced in dollars (dollar pricing). Under dollar pricing, the following are true. First, an identical local good has different prices in two currencies. When the dollar appreciates, identical local goods are more expensive when denominated in US dollars than in the local currency. This generates an inefficient cross-currency price dispersion despite the identical marginal costs of production. Second, since both exports and imports are in dollars, changes in the exchange rate affect import prices and consumption for local households but have a muted effect on consumption for US households.

I find that the optimal FXI responds to this price-dispersion wedge under dollar pricing. By manipulating the exchange rate, FXI affects the relative prices of local goods sold locally and in the US, closing the price dispersion. This makes the optimal FXI particularly responsive to shocks under dollar pricing. Moreover, the transmission channel of FXI is asymmetric across countries. Optimal FXI has a large stabilization effect on local inflation without creating a large US inflationary spillover. These results suggest that dollarization in international trade is a key driver of capital flow stabilization policy in international finance.

Finally, as a robustness check, I deviate from the assumption of international policy coordination. The need to fully specify dynamic games poses many challenges to the modern literature on strategic monetary policy coordination. These include the definitions of equilibria (commitment or discretion), the choice of policy instruments (inflation or output gap), and the feasibility of deriving analytical and numerical solutions. In particular, the model is difficult to solve when there are multiple policy instruments: monetary policy and FXI. [5] Hence, I focus on an analytically tractable case where central banks solely focus on domestic inflation and output

[5] As a step toward characterizing noncooperative strategic interaction between central banks, Benigno and Benigno (2006) and Corsetti et al. (2010) study open-loop Nash equilibrium where central banks in each country specify their state-contingent plans for every possible future state and commit to their sequence of strategies to maximize their objectives. Bodenstein et al. (2019) provides a computational toolbox to solve the open-loop Nash equilibrium, focusing on a case with one policy instrument for each country. Bodenstein et al. (2023) shows that the gains from cooperation increase with the size of asymmetric net foreign asset position. The full characterization of strategic policy interaction in repeated games is beyond the scope of this paper and an interesting avenue for future research.

stabilization but no longer coordinate on international objectives.

As a case study relevant to recent policymakers, I consider the optimal response of FXI by the local central bank to an exogenous increase in US tariffs. This reflects the recent US-China trade war: the US imposes tariffs on imports from China while China counteracts the US by accumulating the dollar reserves to depreciate the yuan and increase export competitiveness. I find that the optimal FXI helps the local monetary authority to stabilize inflation and output but exacerbates the US inflation-output trade-off (beggar-thy-neighbor). Furthermore, this non-cooperative FXI exacerbates distortions to global demand, disproportionately benefiting US households over local households (beggar-thy-self). This analysis provides a rationale for why we need to convince policymakers to coordinate on monetary and exchange rate policies.

Literature. First, this paper builds on models of exchange rate determination in an imperfect financial market (Gabaix and Maggiori 2015, Itskhoki and Mukhin 2021, Maggiori 2022, Fukui et al. 2023). Their models have been used to study the effect of FXI (Fanelli and Straub 2021, Davis et al. 2023, Ottonello et al. 2024).[6] Moreover, Cavallino (2019), Amador et al. (2020), Basu et al. (2020), and Itskhoki and Mukhin (2023) study monetary policy and FXI in a small open economy case where households take the international price and demand as given.[7] My contribution is to study a global implication of FXI in an analytically tractable two-country framework, in which FXI affects international price and demand, and to provide a full analytical characterization of optimal monetary policy and FXI targeting rules. My model suggests a novel trade-off of FXI between internal objectives (inflation and output) and external objectives

[6]The source of non-fundamental exchange rate volatility is modeled in the literature as an exogenous shock to the unhedged carry trade return (Devereux and Engel 2002, Jeanne and Rose 2002, Kollmann 2005) or convenience yield on the dollar bond (Jiang et al. 2021; 2023, Engel and Wu 2023, Kekre and Lenel 2023).

[7]Cavallino (2019) shows that FXI is costly for a central bank since FX purchase lowers the FX return while it is profitable for intermediaries as they take an opposite carry trade position against the central bank. When domestic households do not own the entire share of the intermediaries, FXI trades off the carry cost with exchange rate stabilization. Amador et al. (2020) show that the zero lower bound of the nominal interest rate generates capital inflow since the expected appreciation of local currency is not offset by the lower interest rate. FXI absorbs the capital flows by accumulating foreign reserves but generates a resource cost. Basu et al. (2020) builds an "integrated policy framework" that jointly studies monetary policy, FXI, capital control, and macroprudential regulation. When banks face a sudden outflow of capital, a lower policy rate relaxes the domestic borrowing constraint but tightens the external borrowing constraint due to currency depreciation. FXI limits this depreciation and improves the monetary trade-off. Itskhoki and Mukhin (2023) show that unrestricted use of monetary policy and FXI stabilize both inflation and exchange rate separately. However, when FXI is constrained, monetary policy faces a trade-off between inflation and exchange rate stabilization.

(exchange rate and world demand).[8] Moreover, I take the first step to incorporate FXI in a non-cooperative equilibrium and study the strategic interaction between monetary policy and FXI in a large two-country model.

My paper is also based on the large literature on optimal monetary policy. An early strand of literature studies optimal monetary policy in a small open economy (Clarida et al. 2001, Schmitt-Grohé and Uribe 2001, Kollmann 2002, Galí and Monacelli 2005, Faia and Monacelli 2008). Another strand of papers study international monetary policy transmission and cooperation in a large two-country economy (Obstfeld and Rogoff 2000, Corsetti and Pesenti 2001, Clarida et al. 2002, Benigno and Benigno 2003, Corsetti and Pesenti 2005, Benigno and Benigno 2006, Devereux and Engel 2003, Corsetti et al. 2010; 2020; 2023, Engel 2011). These papers study monetary policy independently of FXI. I contribute to the literature by providing a joint configuration of monetary policy and FXI.

Growing empirical evidence documents that FXI is effective in stabilizing the exchange rate (Dominguez and Frankel 1993, Dominguez 2003, Fatum and Hutchison 2010, Blanchard et al. 2015, Kuersteiner et al. 2018, Adler et al. 2019, Fratzscher et al. 2019, Hofmann et al. 2019, Fratzscher et al. 2023, Rodnyansky et al. 2024). My contribution is to provide a normative analysis based on a full analytical characterization of an optimal FXI targeting rule.

This paper is related to recent literature on the dominance of the US dollar in trade invoicing (Gopinath 2016, Gopinath et al. 2020, Gopinath and Stein 2021, Mukhin 2022, Egorov and Mukhin 2023). My paper bridges the gap between the literature on international trade and finance and suggests a novel mechanism where capital flow management policies are motivated by dollar pricing.

Finally, there are discussions on gains from international monetary policy coordination in a large two-country model (Obstfeld and Rogoff 2002, Benigno and Benigno 2006, Benigno and Woodford 2012, Corsetti et al. 2010, Bodenstein et al. 2023). Fanelli and Straub (2021) and Itskhoki and Mukhin (2023) study international coordination on FXI between small open economies. I take the first step to incorporating monetary policy and FXI in a large two-country framework.

The rest of this paper is organized as follows. Section 2 describes the model environment with

[8] Other complementary transmission channels of FXI can create additional trade-offs. Basu et al. (2020), Davis et al. (2023), Rodnyansky et al. (2024) show that FXI mitigates balance-sheet risk when firms or banks have foreign currency debt. Ottonello et al. (2024) show that FXI is used as an industrial policy and helps the convergence to the technological frontier.

segmented currency markets. Section 3 discusses the intuitive mechanism behind monetary and exchange rate policy interaction. Section 4 formally characterizes optimal monetary policy and FXI under cooperation. Section 5 studies the optimal policies under dollar pricing. Section 6 provides robustness checks. Section 7 concludes.

2 Model Environment: Segmented Currency Markets

The model builds on a standard international real business cycle model (Clarida et al. 2002). The key departure from the literature is that households are restricted from trading bonds denominated in foreign currency. The households' net foreign asset position must be intermediated by global financial intermediaries with limited capacity to bear exchange rate risks. This creates limits to arbitrage opportunities for households, allowing FXI to affect the relative demand and supply of bonds in different currencies (Gabaix and Maggiori 2015, Itskhoki and Mukhin 2021; 2023).

There are two symmetric economies, the US and local (the rest of the world).[9] Each country is populated with a continuum of agents of unit mass. Firms in each country are monopolistic suppliers of one type of tradable good and use labor as the only input in production. Households are allowed to trade state-contingent assets domestically but not internationally. In what follows, I describe the setup focusing on the local economy since similar expressions apply to the US economy. I denote variables related to the US households and firms with an asterisk ($*$) and refer to the US unit of account as the dollar.

Households. In each country, there is a continuum of households that maximize the expected discount value of their lifetime utility. I assume the households have a constant relative risk aversion (CRRA) utility in consumption. The maximization of local households is:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \zeta_l \frac{L_t^{1+\eta}}{1+\eta} \right],$$

where C_t and L_t are the consumption and labor supply, σ is the constant relative risk aversion, β is the discount factor, and η is the inverse Frisch elasticity of labor. The households' consumption

[9] As discussed in Corsetti et al. (2010), the assumption of symmetry allows for an analytically sharp characterization of optimal policy rules. A more general setup would allow for asymmetric degrees in trade openness (home bias in consumption) and country size (different populations).

basket C_t is a constant elasticity of substitution (CES) aggregator of local and US goods:

$$C_t = \left[aC_{Lt}^{\frac{\phi-1}{\phi}} + (1-a)C_{Ut}^{\frac{\phi-1}{\phi}} \right]^{\frac{\phi}{1-\phi}},$$

where C_{Lt} and C_{Ut} are the consumption of local and US goods, a is the weight of the local good, and ϕ is the elasticity of substitution between the local and US goods. I assume $a \in (1/2, 1]$ so that households exhibit a home bias of consumption. In the limiting case where $\sigma = \phi = 1$ (Cole and Obstfeld 1991), households have log and Cobb-Douglas utility. I define $C_t(l)$ as the local households' consumption of a local good h and $C_t(u)$ as their consumption of a US good u. I assume that each good l or u is an imperfect substitute for all other goods' varieties, with constant elasticity of substitution θ:

$$C_{Lt} = \left[\int_0^1 C_t(l)^{\frac{\theta-1}{\theta}} dj \right]^{\frac{\theta}{1-\theta}}, \quad C_{Ut} = \left[\int_0^1 C_t(u)^{\frac{\theta-1}{\theta}} du \right]^{\frac{\theta}{1-\theta}}.$$

The price indices of the local and US goods in terms of the local currency are given by:

$$P_{Lt} = \left[\int_0^1 P_t(l)^{1-\theta} dl \right]^{\frac{1}{1-\theta}}, \quad P_{Ut} = \left[\int_0^1 P_t(u)^{1-\theta} du \right]^{\frac{1}{1-\theta}},$$

and the consumer price index (CPI) associated with the consumption basket C_t is given by:

$$P_t = \left[aP_{Lt}^{1-\phi} + (1-a)P_{Ut}^{1-\phi} \right]^{\frac{1}{1-\phi}}. \tag{1}$$

I assume that local households can only trade local currency bonds. The local households' budget constraint is:

$$P_t C_t + \frac{B_t}{R_t} = B_{t-1} + W_t L_t + \Pi_t + T_t, \tag{2}$$

where B_t is the local households' investment in one-period state non-contingent bonds denominated in the local currency, R_t is the interest rate on the local currency bond ($1/R_t$ is the bond price), W_t is the wage, Π_t is the lump-sum transfer of firms' profit, and T_t is the government transfer.

The households' Euler equation and labor supply equation are:

$$1 = \beta R_t E_t \left(\frac{C_{t+1}}{C_t}\right)^{-\sigma} \frac{P_t}{P_{t+1}},$$
$$\frac{W_t}{P_t} = C_t^{\sigma} L_t^{\eta}.$$

The households' intratemporal consumption allocation problem gives the following demand for local and US goods:

$$C_{Lt} = a\left(\frac{P_{Lt}}{P_t}\right)^{-\phi} C_t, \quad C_{Ut} = (1-a)\left(\frac{P_{Ut}}{P_t}\right)^{-\phi} C_t,$$

and the demand for differentiated goods produced within each country:

$$C_t(l) = \left(\frac{P_t(l)}{P_{Lt}}\right)^{-\theta} C_{Lt}, \quad C_t(u) = \left(\frac{P_t(u)}{P_{Ut}}\right)^{-\theta} C_{Ut}.$$

The US demands and prices are defined similarly. The CPI faced by US households is:

$$P_t^* = \left[a {P_{Lt}^*}^{1-\phi} + (1-a){P_{Ut}^*}^{1-\phi}\right]^{\frac{1}{1-\phi}},$$

where P_{Lt}^* and P_{Ut}^* are the prices of local and US goods in terms of US dollars.

Let $\mathcal{E}_t$ denote the nominal exchange rate in terms of the local unit of account per dollar. The real exchange rate is defined as the ratio of CPIs expressed in the same currency, i.e., $e_t \equiv \mathcal{E}_t P_t^*/P_t$. An increase in e_t corresponds to a real depreciation, that is, a decrease in the price of the local households' consumption basket relative to that in the US. The terms of trade are defined as the relative price of local imports in terms of exports: $\mathcal{T}_t \equiv P_{Ut}/\mathcal{E}_t P_{Lt}^*$.

Firms. Firms use domestic labor to produce a differentiated good l following a production function:

$$Y_t(l) = A_t L_t(l),$$

where $Y_t(l)$ is the output and $L_t(l)$ is the labor input for the producer of good l. A_t is a technology shock common to all firms and follows an AR(1) process: $\log(A_t) = \rho_a \log(A_{t-1}) + \sigma_a \epsilon_{at}$, where ρ_a and σ_a are the persistence and the standard deviation, respectively. Let Y_{Lt} =

$\left[\int_0^1 Y_t(l)^{\frac{\theta-1}{\theta}} dl\right]^{\frac{\theta}{\theta-1}}$ be the final output of the local good. The demand for the differentiated good l is given by:

$$Y_t(l) = \left(\frac{P_t(l)}{P_{Lt}}\right)^{-\theta} Y_{Lt}.$$

Firms are subject to nominal rigidity (Calvo 1983) so that, in each period, firms update their prices with probability ξ_p. To capture the key intuition, I assume producer currency pricing (PCP) so that the export price is sticky in the exporters' currency (Section 5 derives the optimal policy under dollar pricing). The firms' maximization problem is:

$$\max_{\{P_t(l), \mathcal{E}_t P_t^*(l)\}} E_t \left\{ \sum_{k=0}^{\infty} Q_{t,t+k} \xi_p^k \begin{pmatrix} (1+s_t)\left[P_t(l)Y_{t+k}(l) + \mathcal{E}_t P_t^*(l) Y_{t+k}^*(l)\right] \\ -MC_{t+k}(l)\left[Y_{t+k}(l) + Y_{t+k}^*(l)\right] \end{pmatrix} \right\}, \tag{3}$$

where $Q_{t,t+k} = \beta^t \left(\frac{C_{t+k}}{C_t}\right)^{-\sigma} \frac{P_t}{P_{t+k}}$ is the households' stochastic discount factor, $Y_t(l)$ and $Y_t^*(l)$ are demand for local goods by local and US households, $P_t(l)$ is the price of local goods in local currency charged to local households, $P_t^*(l)$ is the price of local goods in dollars charged to local and US households, $MC_t = W_t/P_t A_t$ is the marginal cost of production, and s_t is the sales subsidy.

Solving the maximization problem, the optimal price of local goods charged to local households is:

$$P_t(l) = \frac{\theta}{(\theta-1)(1+s_t)} \frac{E_t \sum_{k=0}^{\infty} Q_{t,t+k} \xi_p^k Y_{t+k}(l) MC_{t+k}}{E_t \sum_{k=0}^{\infty} Q_{t,t+k} \xi_p^k Y_{t+k}(l)}.$$

The law of one price (LOOP) implies that local and US households are charged an identical price for local goods when expressed in the same currency, i.e., $P_t^*(l) = P_t(l)/\mathcal{E}_t$.

Assuming symmetry so that all firms that can update the prices choose the same value, I obtain the law of motion of local good prices faced by local households:

$$P_{Lt}^{1-\theta} = \xi_p P_{Lt-1}^{1-\theta} + (1-\xi_p) P_t(l)^{1-\theta}.$$

International Financial Markets. To study the effects of monetary policy and FXI jointly, I make a key departure from the standard international business cycle model by introducing limits to arbitrage in the international financial market. There is a well-known empirical fact that unhedged rates of returns across currencies are not equalized. In other words, the uncovered interest rate parity (UIP) condition does not hold (Fama 1984). Motivated by this fact, I assume that households are restricted from trading foreign currency assets: otherwise, households would gain infinite carry-trade profits by investing in high-return currencies (Jeanne and Rose 2002, Gabaix and Maggiori 2015, Itskhoki and Mukhin 2021). Under limits to arbitrage, FXI can affect the exchange rate by changing the relative demand and supply of two currencies.

To give a simple example, suppose that the local central bank buys the local currency and sells US dollars. This appreciates the local currency and reduces the return on investment in the local currency relative to the dollar. However, despite this rate of return differential, households cannot take advantage of this arbitrage opportunity by borrowing in the local currency and investing in US dollars. Hence, FXI increases the demand for the local currency relative to the dollar and appreciates the local currency.

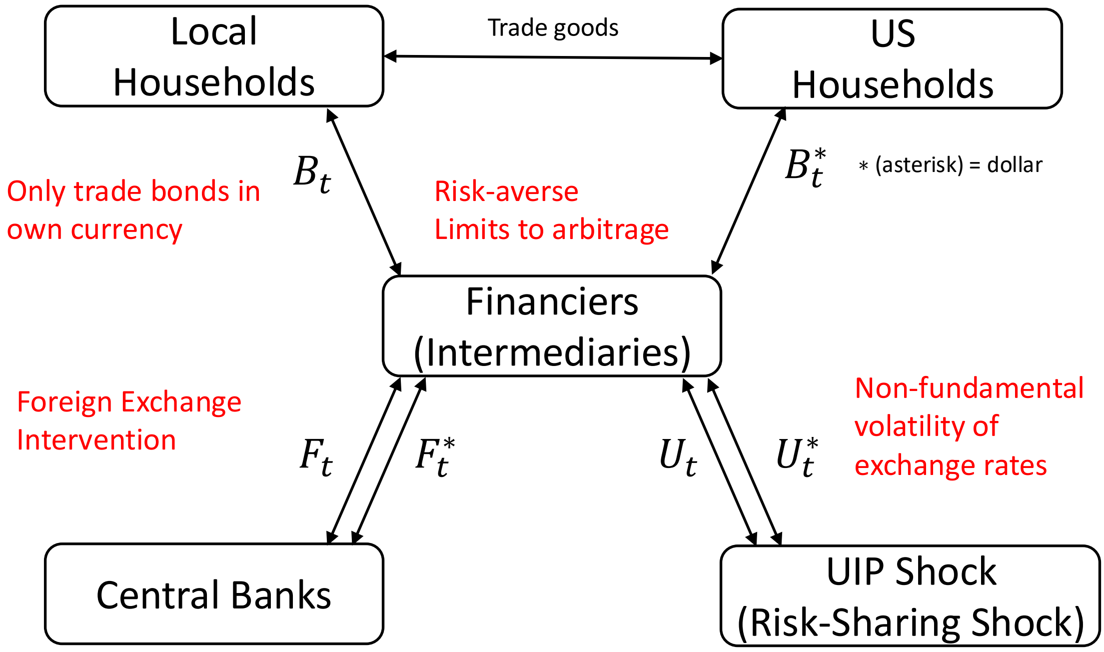
More formally, this financial imperfection is modeled by introducing limits to international financial intermediation. Figure 1 shows the basic structure of the international financial market. Households can only trade bonds in their own currency and their net foreign asset position must be intermediated by financiers (global financial intermediaries) who have limited capacity to bear exchange rate risks.[10] The central banks in the two countries have conventional and unconventional policy instruments: they use monetary policy to set nominal interest rates and FXI to buy and sell bonds in two currencies.[11] Finally, I introduce an exogenous shock to the UIP deviation, which is a common way in the literature to generate high exchange rate volatility and lack of correlation between exchange rates and macroeconomic fundamentals (exchange rate disconnect). The UIP shock can be understood as an exogenous demand for dollar bonds due to their liquidity and safety of dollar bonds.[12]

[10]To characterize the optimal policy analytically, I assume that households cannot access foreign currency bonds, following Gabaix and Maggiori (2015) and Itskhoki and Mukhin (2021). Fukui et al. (2023) generalize this setup so that households and firms can trade in both domestic and foreign currencies but it is costly to access foreign currency bonds.

[11]Under cooperation, whether the local or US central bank uses FXI does not affect the key results since the relative demand and supply of bonds held by the two central banks determine the exchange rate.

[12]Since the focus of this paper is the economic consequence of UIP shocks and the role of monetary and exchange rate policies, the model is agnostic about the source of UIP shocks to keep tractability. There is an extensive discussion on the drivers of UIP shocks, including investors' heterogeneous beliefs (Bacchetta and Van Wincoop 2006) and cognitive bias (Burnside et al. 2011), rare disaster risk (Farhi and Gabaix 2016), interbank friction (Bianchi et al. 2023a), and special role of US Treasury bonds, such as liquidity or collateral values.(Bianchi et al. 2023a;b). How different sources of UIP shocks affect the optimal policy design is beyond the scope of this paper and is left for future research.

Figure 1: Basic Structure of the International Financial Market



Note: The figure shows the basic structure of the international financial market. Local and US households can only trade bonds in their own currency (B_t, B_t^*). The local central bank uses foreign exchange intervention to trade bonds in two currencies (F_t, F_t^*). The liquidity traders generate an exogenous UIP shock (U_t, U_t^*). Financiers (global financial intermediaries) intermediate the net foreign asset positions of the households, the local central bank, and the liquidity traders.

I will formalize this intuition in the model. There is a measure m_u of liquidity traders who generate an exogenous UIP shock. The investors hold a zero-net portfolio (U_t, U_t^*) so the investment U_t^* in dollar bonds is matched by the investment $U_t/R_t = -\mathcal{E}_t U_t^*/R_t^*$ in the local currency bonds. The positive U_t^* implies that the liquidity traders borrow in the local currency and lend in dollars, and vice versa. I assume that the liquidity traders' position follows an AR(1) process: $U_t = \rho_u U_{t-1} + \sigma_u \epsilon_{ut}$.

Central banks use sterilized intervention: if they buy dollar bonds, they sell an equivalent volume of local currency bonds and vice versa. The local central bank holds a zero-net portfolio (F_t, F_t^*) given by $F_t/R_t = -\mathcal{E}_t F_t^*/R_t^*$ and its profits and losses are transferred to the local households in a lump-sum way.[13]

There is a measure m_d of financiers who intermediate the portfolio positions of the households, liquidity traders, and central banks. The financiers hold a zero-net portfolio (D_t, D_t^*)

[13] I assume that FXI is unconstrained for simplicity. In reality, central banks face a zero lower bound on FX reserves, which creates an additional policy trade-off. Davis et al. (2023) show that, when reserves cannot be borrowed, the optimal policy is to accumulate the FX reserves during normal times and sell them during crisis times.

given by $D_t/R_t = -\mathcal{E}_t D_t^*/R_t^*$. Following Itskhoki and Mukhin (2021) and Fukui et al. (2023), I assume that the financiers maximize the following constant absolute risk aversion (CARA) utility:

$$\max_{d_t} E_t \left\{ -\frac{1}{\omega^F} \exp\left(-\omega^F \bar{R}_t d_t \right) \right\}, \tag{4}$$

where $d \equiv D_t/P_t$ is the real investment in local currency bonds, $\omega^F \geq 0$ is a risk aversion of financiers, and

$$\bar{R}_t \equiv R_t - R_t^* \frac{\mathcal{E}_{t+1}}{\mathcal{E}_t}$$

is the unhedged return on the carry trade.[14][15] In a limiting case where $\omega^F = 0$, arbitrageurs are risk-neutral and trade both currencies without charging a risk premium. Hence, the UIP holds and the expected excess return is zero ($E_t \bar{R}_t = 0$). However, when $\omega^F > 0$, arbitrageurs are risk-averse and require a risk premium for trading bonds in two currencies, which drives the UIP deviation ($E_t \bar{R}_t \neq 0$).

The market clearing conditions for the bond market imply the net demands for local currency and dollar bonds are zero:

$$B_t + U_t + D_t + F_t = 0, \quad \text{and} \quad B_t^* + U_t^* + D_t^* + F_t^* = 0. \tag{5}$$

The competitive equilibrium is defined as the set of prices, quantities, and policy variables that solve the maximization problems of households, firms, and financiers under the constraints and market clearing conditions.

[14]As discussed in Itskhoki and Mukhin (2021), the assumption of a CARA utility improves the traceability since their portfolio decision does not depend on the intermediaries' wealth, allowing us to avoid an additional state variable. The limitation of this setting is that I assume banks maximize their one-period return. Potential extensions to microfound banks' risk-aversion include introducing occasionally binding borrowing constraints, costs of currency hedging, or liquidity holdings by banks (Bianchi et al. 2023a). These extensions would allow for more realistic trade-offs between risks and returns in a dynamic optimization problem.

[15]I assume that the financiers' profit is transferred to the local households as a lump-sum payment. As discussed in Appendix A.2, the profits and losses generated by carry trade positions do not affect the first-order dynamics of the model.

3 International Risk-Sharing and Policy Trade-offs

This section studies how FXI mitigates the monetary policy trade-off between inflation and output. I construct an intuition step-by-step to understand the key policy trade-off in a two-country economy. In subsection 3.1, I briefly describe the already well-established insight in the literature (Corsetti et al. 2023) as a starting point. I define the concept of international risk-sharing and study how it generates a monetary policy trade-off. Next, subsection 3.2 illustrates the key contribution of my paper by studying how FXI mitigates the monetary policy trade-off by affecting international risk-sharing.

3.1 Starting point: Monetary Policy Trade-off due to the Lack of Risk-Sharing

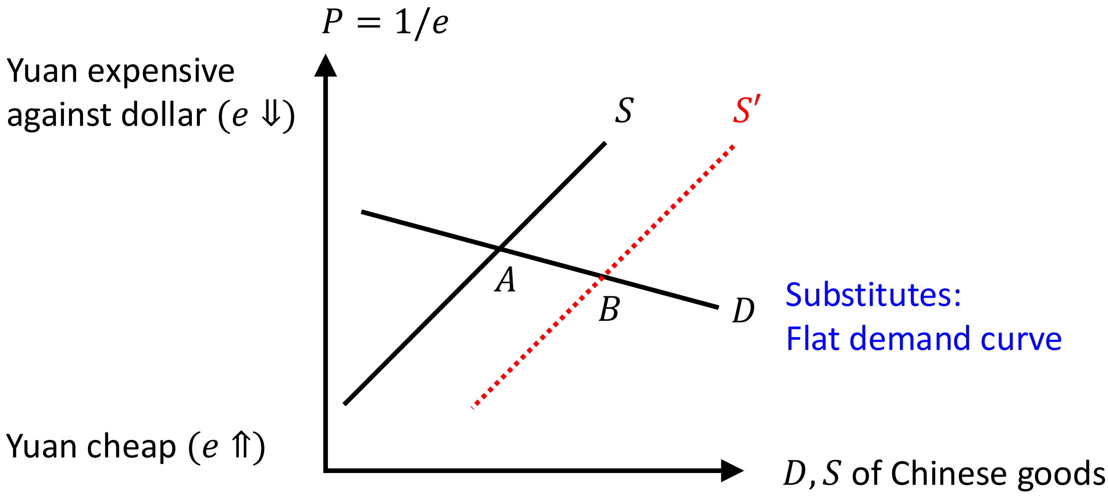
To capture intuition, I explain how a lack of international risk-sharing generates a monetary policy trade-off between inflation and output, which is already well-studied in the literature (Corsetti et al. 2023). This is important since, as discussed in Section 4, under cooperation, central banks target risk-sharing in addition to inflation and output.

International Risk-Sharing: Concept and Definition. International risk-sharing implies that the exchange rate adjusts to smooth consumption across countries. To understand the concept, consider an example of two countries: China (the local economy) and the US. Figure 2 plots the demand and supply curves for Chinese goods. The x-axis shows the demand and supply of Chinese goods and the y-axis shows the real exchange rate or the relative prices of Chinese and US goods. Suppose that productivity increases in China, which moves the supply curve to the right from S to S' and increases the consumption in China from point A to B. At the same time, the price of Chinese goods decreases and the yuan depreciates (becomes cheaper) against the dollar. This implies that US households face lower import prices and US consumption increases. Hence, both local and US households increase consumption, even if they cannot trade state-contingent assets internationally.

Formally, I define the risk-sharing wedge as the difference in marginal utilities of consumption between the two countries:

$$\mathcal{W}_t \equiv \frac{U'(C_t^*)}{U'(C_t)}\frac{1}{e_t} = \frac{\left(C_t^*\right)^{-\sigma}/\mathcal{E}_t P_t^*}{C_t^{-\sigma}/P_t} = \left(\frac{C_t}{C_t^*}\right)^{\sigma}\frac{1}{e_t}.$$

Figure 2: International Risk-Sharing: Concept



Note: This figure provides a graphical illustration of international risk-sharing between China and US households. The x-axis shows demand and supply of Chinese goods and the y-axis shows the real exchange rate.

I define the variables with tilde as deviations from their first-best allocation values. In a log-linearized form, the risk-sharing wedge can be expressed as:

$$\tilde{\mathcal{W}}_t \equiv \sigma(\tilde{C}_t - \tilde{C}_t^*) - \tilde{e}_t. \tag{6}$$

Equation (6) implies that the risk-sharing wedge $\tilde{\mathcal{W}}_t$ is positive when local consumption is higher than US consumption ($\tilde{C}_t > \tilde{C}_t^*$) or the local currency appreciates against the dollar (lower $\tilde{e}_t$) so that local households face lower import prices and increase consumption.

Whether international risk-sharing holds or not depends on the elasticity of substitution between local and US goods. Figure 2 focuses on the case where local and US goods are substitutes so that the demand curve is flat. When productivity in China increases and the supply curve shifts from S to S', consumption C_t in China increases by a large degree but the Chinese yuan depreciates (e_t increases) by a small degree. As a result, a positive productivity shock generates an excess demand for Chinese households, making the risk-sharing wedge $\tilde{\mathcal{W}}_t$ positive.[16]

[16] Conversely, when the local and US goods are complements, the risk-sharing wedge becomes negative when productivity in China increases. In a special case where households have a log and Cobb-Douglas utility so that local and US goods are neither substitutes nor complements, productivity shocks do not generate the risk-sharing wedge (Cole and Obstfeld 1991). In this special case, there is no trade-off between inflation and output. I focus on the case where goods are substitutes based on standard calibration in the international business cycle literature, as discussed in Section 4.

As discussed in Corsetti et al. (2008), this intuition can be confirmed analytically in a special case of log utility and financial autarky: an extreme form of market incompleteness where no international financial transaction is allowed. Under financial autarky, trade must be balanced in each period so that:

$$P_{Ut}C_{Ut} = e_t P^*_{Lt} C^*_{Lt}.$$

In this case, the risk-sharing wedge can be written in terms of the output in the two countries:

$$\tilde{\mathcal{W}}_t = \frac{2a(\phi - 1)}{1 - 2a(1 - \phi)}(\hat{A}_t - \hat{A}^*_t), \tag{7}$$

where A_t and A^*_t are percentage deviations of local and US productivity from the steady state, respectively. Equation (7) shows that when the local and US goods are substitutes ($\phi > 1$), a positive supply shock in the local economy generates local excess demand and makes the risk-sharing wedge positive. Although this is an extreme assumption for analytical tractability, this intuition can be confirmed in a more general case where households trade state non-contingent bonds (Section 4).

This lack of risk-sharing is a crucial driver of inflation and creates a non-trivial trade-off of monetary policy between inflation and output. I define $\pi_{Lt} = P_{Lt}/P_{Lt-1} - 1$ as the inflation rate of locally-produced goods consumed by local households and $\pi^*_{Ut} = P^*_{Ut}/P^*_{Ut-1} - 1$ as the inflation rate of US-produced goods consumed by US households (producer-price inflation). Log-linearization of Equation (3) shows that the New Keynesian Phillips Curves (NKPCs) can be written in terms of the price misalignment and risk-sharing wedge:

$$\pi_{Lt} = \beta E_t \pi_{Lt+1} + \kappa\left[(\sigma + \eta)\tilde{Y}_{Lt} - 2a(1 - a)(\sigma\phi - 1)\tilde{\mathcal{T}}_t + (1 - a)\tilde{\mathcal{W}}_t + \mu_t\right]. \tag{8}$$

$$\pi^*_{Ut} = \beta E_t \pi^*_{Ut+1} + \kappa\left[(\sigma + \eta)\tilde{Y}_{Ut} + 2a(1 - a)(\sigma\phi - 1)\tilde{\mathcal{T}}_t - (1 - a)\tilde{\mathcal{W}}_t + \mu^*_t\right]. \tag{9}$$

Equation (8) implies that, as in the standard New Keynesian model in a closed economy, current inflation depends on future expected inflation and the output gap, defined as the deviation of output from the first-best efficient level: $\tilde{Y}_{Lt} \equiv \hat{Y}_{Lt} - \hat{Y}^{fb}_{Lt}$. In a two-country economy, inflation also depends on the terms-of-trade gap $\tilde{\mathcal{T}}_t \equiv \hat{\mathcal{T}}_t - \hat{\mathcal{T}}^{fb}_t$ and risk-sharing wedge $\tilde{\mathcal{W}}_t$. The terms-of-trade gap is defined as the deviation of terms of trade from the efficient level. When the terms-of-trade gap is negative (the import price is lower than the export price) and the trade elasticity ϕ is sufficiently high, local households face lower import prices and increase consumption. This increases the marginal cost of production and increases inflation.[17] When the risk-sharing wedge is positive (local households have excess demand), inflation increases due to a higher marginal cost. Finally, μ_t is the markup shock, which drives the cost-push inflation.

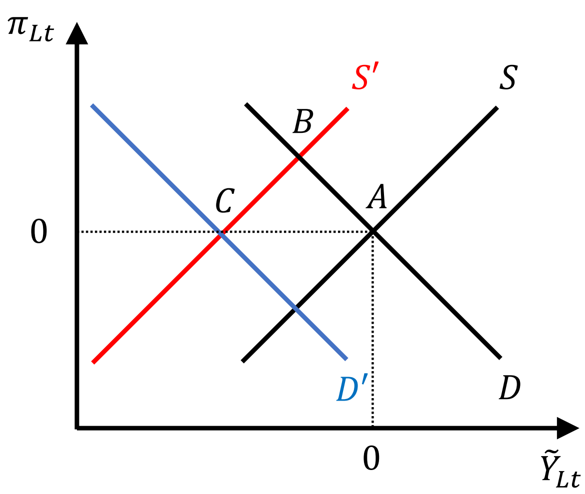
To intuitively understand the mechanism where a lack of risk-sharing generates the inflation-output trade-off of monetary policy, consider a special case where monetary policy strictly commits to targeting zero inflation rate of locally produced goods ($\pi_{Lt} = 0$). Figure 3 plots the demand and supply curves for the local goods. I take the local output gap $\tilde{Y}_{Lt}$ on the x-axis and inflation π_{Lt} on the y-axis. Consider an increase in the local productivity A_t, which affects inflation via two channels. The first channel is a direct effect similar to the closed economy New Keynesian model. Since a positive supply shock generates deflation, the optimal policy is to reduce the interest rate so that both inflation and the output gap equal zero (point A). The second channel is an indirect effect via risk-sharing unique to the two-country economy. As shown in Equation (7), an increase in local supply generates local excess demand and a positive demand gap. As implied by Equation (8), this positive demand gap, in turn, increases the marginal cost of production and inflation rate of local goods. The supply curve shifts from S to S' and inflation increases from point A to B. Due to strict inflation targeting, the central bank increases the interest rate and reduces the demand. The demand curve shifts from D to D', which stabilizes inflation but makes the output gap negative (point C). Hence, stabilizing inflation comes at the cost of destabilizing the output gap due to the lack of risk-sharing.

3.2 Mechanism: Interdependence of Monetary Policy and FXI

The previous subsection studied the monetary policy trade-off due to the lack of risk-sharing. This section discusses the key mechanism where FXI can mitigate this monetary policy trade-off by affecting the risk-sharing. First, I characterize the relationship between the risk-sharing, UIP deviation, and FXI, as studied in Gabaix and Maggiori (2015) and Itskhoki and Mukhin (2023).

[17] As discussed in Clarida et al. (2002), the effect of the terms-of-trade gap on inflation depends on the trade elasticity. Consider an increase in the US output, which depreciates the US dollar and decreases the import price of US goods faced by local households. On the one hand, lower import prices increase consumption demand by local households. This increases the marginal cost of production and inflation. On the other hand, higher export prices for local firms increase the marginal benefit of production for local firms and reduce inflation. When local and US goods are substitutes ($\sigma\phi > 1$), the former effect dominates the latter, so π_{Lt} is decreasing in $\tilde{\mathcal{T}}_t$.

Figure 3: Monetary Policy Trade-off due to the Lack of International Risk-Sharing



Note: This figure illustrates how an increase in local productivity generates a monetary policy trade-off due to the lack of international risk-sharing. The x- and y-axes show the output gap and inflation rate of local goods, respectively.

Next, I explain intuitively the interdependence of monetary policy and FXI, which is the key contribution of my paper. This intuition will be formalized as a solution to the optimal policy problem in Section 4.

The Effect of FXI on the International Risk-Sharing. This section characterizes the equilibrium relationship between the risk-sharing wedge, UIP deviation, and FXI. Combining the solutions to households' and financiers' problems gives the following lemma:

Lemma 1 (Itskhoki and Mukhin (2021))**.** *The equilibrium condition in the financial market, which is log-linearized under a symmetric steady state, can be written as:*

$$E_t\tilde{\mathcal{W}}_{t+1} - \tilde{\mathcal{W}}_t = \tilde{r}_t - \tilde{r}_t^* - (E_t\tilde{e}_{t+1} - \tilde{e}_t) = \omega(f_t^* - u_t) + \omega_b b_t^*, \tag{10}$$

where $r_t \equiv R_t - E_t\pi_{t+1}$, $r_t^* \equiv R_t^* - E_t\pi_{t+1}^*$, $f_t^* \equiv F_t^*/\overline{Y}$, $u_t \equiv U_t/\overline{Y}$, $b_t \equiv B_t/\overline{Y}$, $\omega \equiv m_u(\omega^F\sigma_{et}^2/m_d)$, *and* $\omega_b \equiv \overline{Y}(\omega^F\sigma_{et}^2/m_d)$ *for finite* $\omega\sigma_{et}^2/m_d$, *where* $\overline{Y} \equiv \overline{Y}_L = \overline{Y}_U$ *is GDP under the symmetric steady state and* $\sigma_{et}^2 \equiv var(\Delta\log\mathcal{E}_{t+1})$ *is the standard deviation of the change in log nominal exchange rate* $(\Delta\log\mathcal{E}_{t+1} \equiv \log\mathcal{E}_{t+1} - \log\mathcal{E}_t)$.

Proof. See Appendix A.2.

Intuitively, consider the limiting case where $\omega, \omega_b \to 0$ so that financiers can arbitrage away any excess returns across currencies. In this case, the interest rate differential across currencies is offset by expected exchange rate depreciation, and the UIP condition holds. Since households in the two countries face the same rates of return on investment, they have the same incentive to consume and save. As a result, the risk-sharing wedge is expected to be unchanged over time; that is, consumption smoothing holds on average. However, as studied in subsection 3.1, this does not necessarily imply that the risk-sharing wedge is zero for every state of the economy. When local and US goods are substitutes, the risk-sharing wedge is non-zero in general, which creates a monetary policy trade-off between inflation and output.

However, when financiers are risk-averse ($\omega, \omega_b > 0$), the risk-sharing wedge and UIP deviation are determined by the relative demand for domestic and foreign currencies by the central bank, liquidity trader, and households. Suppose that the central banks buy dollars and sell the local currency using FXI ($f_t^* > 0$), which appreciates the dollar relative to the local currency ($\tilde{e}_t$ increases). Since it is costly to invest in dollars, the return on investing in the local currency is higher than the dollar. However, when financiers have limited capacity to bear exchange rate risks ($\omega > 0$), they cannot take full advantage of arbitrage opportunities by investing in the local currency and borrowing in dollars. Hence, a dollar purchase appreciates the dollar and makes the UIP deviation positive. Since the local currency has a higher return than the dollar, local households have a higher incentive to save and a lower incentive to consume. Furthermore, an appreciation of the US dollar increases the import prices faced by local households and reduces their income. Hence, local households' excess demand decreases, and the risk-sharing wedge becomes more negative. Similarly, a demand for the local currency bonds by liquidity traders ($u_t > 0$) makes the UIP deviation negative, and a demand for the dollar bonds by households ($b_t^* > 0$) makes it positive.

FXI Mitigates the Monetary Policy Trade-off. Having discussed how FXI affects the exchange rate and risk-sharing wedge, this section shows a mechanism where FXI affects the inflation-output trade-off of monetary policy, which is the main contribution of my paper.

To capture the intuition, consider the simplest case where the central bank strictly targets zero inflation of producer prices ($\pi_{Lt} = 0$ for all t). Assuming no markup shocks, we can rewrite Equation (8) so that the output gap depends positively on the terms-of-trade gap and negatively on the risk-sharing wedge:

$$(\sigma + \eta)\tilde{Y}_{Lt} = 2a(1 - a)(\sigma\phi - 1)\tilde{\mathcal{T}}_t - (1 - a)\tilde{\mathcal{W}}_t. \tag{11}$$

FXI affects the inflation-output trade-off of monetary policy via changes in the risk-sharing wedge (income effect) and the terms-of-trade gap (substitution effect). To illustrate the mechanism, Figure 4 plots the demand and supply curves for local goods. Intuitively, suppose that central banks purchase dollars and sell the local currency so that the dollar appreciates. First, the income effect implies that when the dollar appreciates, local households face higher import prices and lower income. This decreases the local households' excess demand $\tilde{\mathcal{W}}_t$ and lowers the marginal cost of production. As implied by the NKPC (Equation (8)), this lower marginal cost shifts the supply curve from S to S' and the equilibrium from point A to B. Due to strict inflation targeting, central banks stabilize inflation by lowering the interest rate, which shifts the demand curve from D to D' and the equilibrium from point B to C. Hence, a dollar purchase makes the output gap more positive.

Second, the substitution effect implies that households shift their demand between local and US goods when FXI changes the exchange rate. The equilibrium relationship between the real exchange rate and terms of trade can be written as:

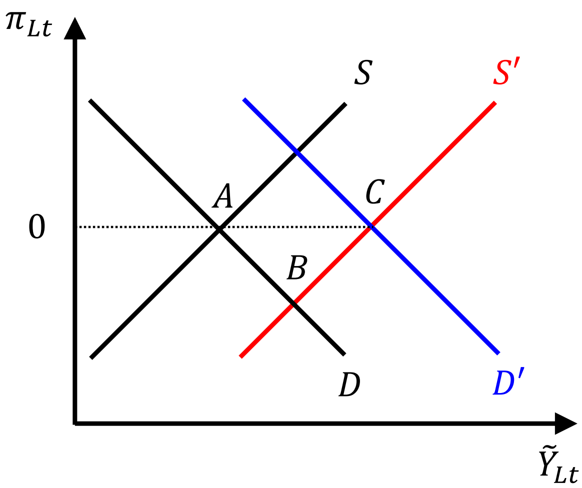
$$\tilde{e}_t = (2a - 1)\tilde{\mathcal{T}}_t, \tag{12}$$

where $2a - 1 > 0$ under the home bias of consumption. When central banks buy the dollar, the dollar appreciates ($\tilde{e}_t$ increases) and the price of US goods increases ($\tilde{\mathcal{T}}_t$ increases). Hence, households shift their demand from the US to local goods. This shifts the demand curve to the right and makes the output gap more positive.

The key insight from this analysis is that the social planner cares about both domestic distortions (inflation and output) and global distortion (risk-sharing) since the risk-sharing or the relative demand across countries is a crucial determinant of inflation and the output gap. FXI affects the inflation-output trade-off of monetary policy via changes in the exchange rate and global demand for goods in different countries and currencies. This intuition will be formally characterized as the solution to the central banks' optimal policy problem in Section 4.

This paper focuses on the complementary role of FXI to monetary policy since the income and substitution effects of FXI have fundamentally different implications from productivity or

Figure 4: FXI Affects the Monetary Policy Trade-off



Note: This figure illustrates how a purchase of US dollars using FXI affects the monetary policy trade-off. The x- and y-axes show the output gap and inflation rate of local goods, respectively.

monetary policy shock in standard New Keynesian models. This difference can be understood analogously to the different transmission channels of productivity and UIP shocks discussed by Itskhoki and Mukhin (2021). A positive productivity shock implies that consumption is high when prices are low. However, FXI implies the opposite: consumption is low when prices are low. When a US dollar purchase depreciates the local currency, local consumption decreases due to higher import prices (income effect), while the relative demand for local goods and local output increase since the local currency is cheap (substitution effect), generating a negative correlation between consumption and output in contrast to standard models. My contribution to the literature is to study the interdependence between monetary policy and FXI. My paper focuses on how these income and substitution effects of FXI mitigate the monetary policy trade-off and formally characterize the optimal FXI rule and its transmission channels.

4 Optimal Trade-offs of Monetary Policy and FXI

Having studied the mechanism behind the interplay of monetary and exchange rate policies, this section provides a formal characterization of optimal policy rules and shows that the optimal FXI is to mitigate the monetary policy trade-off between inflation and output. In subsection 4.1, I provide an analytical characterization of optimal monetary policy and FXI rules in a two-country economy, which is the key contribution of my paper. In subsection 4.2, to show that

FXI has quantitatively large effects, I calibrate the model and compare the transmission of shocks when FXI is available and not available.

4.1 Optimal Policy Rules and Shock Transmission: Analytical Results

Assuming cooperation and commitment, the global social planner sets the nominal interest rate and the volume of FXI to maximize the sum of the welfare of representative households in the two countries. As shown in Corsetti et al. (2023), under PCP, this maximization problem is equivalent to minimizing the following loss function:

$$\mathcal{L} = -E_0 \sum_{t=0}^{\infty} \beta^t \frac{1}{2} \left[\begin{array}{l} \frac{\theta}{\kappa}\left(\pi_{Lt}^2 + \pi_{Ut}^{*2}\right) + (\sigma+\eta)\left(\tilde{Y}_{Lt}^2 + \tilde{Y}_{Ut}^2\right) \\ -\frac{2a(1-a)(\sigma\phi-1)\sigma}{4a(1-a)(\sigma\phi-1)+1}\left(\tilde{Y}_{Lt} - \tilde{Y}_{Ut}\right)^2 + \frac{2a(1-a)\phi}{4a(1-a)(\sigma\phi-1)+1}\tilde{\mathcal{W}}_t^2 \end{array} \right], \tag{13}$$

where:

$$\tilde{Y}_{Lt} - \tilde{Y}_{Ut} = \sigma^{-1}[\{4a(1-a)(\sigma\phi-1)+1\}\tilde{\mathcal{T}}_t + (2a-1)\tilde{\mathcal{W}}_t]. \tag{14}$$

This implies that the planner minimizes the weighted sum of producer-price inflation in each country, the output gap in each country, the terms-of-trade gap, and the risk-sharing wedge across countries.

Following Itskhoki and Mukhin (2021), to focus on the role of financial sectors in driving the UIP deviation, hereafter I consider the limiting case where $\omega_b = 0$, so that:[18]

$$E_t\tilde{\mathcal{W}}_{t+1} - \tilde{\mathcal{W}}_t = \tilde{r}_t - \tilde{r}_t^* - (E_t\tilde{e}_{t+1} - \tilde{e}_t) = \omega(f_t^* - u_t). \tag{15}$$

This assumption allows for a sharp analytical characterization of the optimal FXI rule.

I consider two cases to build up the intuition step-by-step. In the first case, monetary policy in the two countries strictly targets zero producer price inflation, and only FXI is set optimally. In the second case, both monetary policy and FXI are set optimally.

[18] This assumption can be interpreted so that the size of the financial sector, including liquidity traders (m_u) and financiers (m_d), are sufficiently large relative to the real sector.

4.1.1 Case 1: Optimal FXI under Inflation-Targeting Monetary Policy

First, I consider a special case where central banks in the two countries set the nominal interest rate to target zero producer-price inflation. This allows for sharp intuition and analytical characterization of shock transmission channels. The following proposition characterizes the optimal FXI rule and the transmission of productivity shocks, comparing the cases with and without FXI. The transmission channels of FXI can be characterized analytically under the special case where households are not allowed to trade assets internationally (financial autarky) and where households have log utility. However, subsection 4.2 shows that this result also holds in a more general case where households can trade assets internationally with financiers.

Proposition 1 (Optimal FXI under Inflation-Targeting Monetary Policy)**.** *Assume PCP, cooperation, and commitment, and consider the case where monetary policy targets $\pi_{Lt} = \pi^*_{Ut} = 0$ for all t.*

(i) The optimal FXI rule is characterized as:

$$f_t^* = -\xi_Y\{(\tilde{Y}_{Lt} - E_t\tilde{Y}_{Lt+1}) - (\tilde{Y}_{Ut} - E_t\tilde{Y}_{Ut+1})\} + u_t, \tag{16}$$

where $\xi_Y > 0$ is a constant given in the appendix.

(ii) Assume financial autarky and consider the case where local and US goods are substitutes. In response to an increase in local productivity, the optimal FXI is to buy the US dollar and sell the local currency:

$$\frac{\partial f_t^*}{\partial A_t} > 0.$$

Comparing the responses of the output gaps and the risk-sharing wedge to a productivity shock when FXI is available and not available,

$$\left(\frac{\partial \tilde{Y}_{Lt}}{\partial \hat{A}_t}\right)^{No\ FXI} < 0, \qquad \left(\frac{\partial \tilde{Y}_{Lt}}{\partial \hat{A}_t}\right)^{No\ FXI} < \left(\frac{\partial \tilde{Y}_{Lt}}{\partial \hat{A}_t}\right)^{FXI}, \tag{17}$$

$$\left(\frac{\partial \tilde{Y}_{Ut}}{\partial \hat{A}_t}\right)^{No\ FXI} > 0, \qquad \left(\frac{\partial \tilde{Y}_{Ut}}{\partial \hat{A}_t}\right)^{No\ FXI} > \left(\frac{\partial \tilde{Y}_{Ut}}{\partial \hat{A}_t}\right)^{FXI}, \tag{18}$$

and

$$\left(\frac{\partial \tilde{\mathcal{W}}_t}{\partial A_t}\right)^{No\ FXI} > 0, \qquad \left(\frac{\partial \tilde{\mathcal{W}}_t}{\partial A_t}\right)^{No\ FXI} > \left(\frac{\partial \tilde{\mathcal{W}}_t}{\partial A_t}\right)^{FXI}. \tag{19}$$

Proof. See Appendix A.3.

Part (i) of Proposition 1 shows that the optimal FXI can be written as a function of the relative output gap growth across the two countries and the UIP shock. First, the optimal FXI is to buy the US dollar ($f_t^* > 0$) when the local output gap is negative ($\tilde{Y}_{Lt} < 0$) or the US output gap is positive ($\tilde{Y}_{Ut} > 0$). Second, when there is an exogenous increase in liquidity traders' demand for the local currency, the optimal FXI is to offset this local currency demand by buying the US dollar.[19]

Part (ii) shows that, in response to an increase in local productivity, the optimal FXI is to buy the US dollar. This mitigates the on-impact response of the output gap and risk-sharing wedge to a productivity shock. When FXI is available, the local (US) output gap is less negative (positive) and the risk-sharing wedge is less positive than in the case where FXI is unavailable.[20]

Proposition 1 provides a mathematical formalization of intuition obtained in Section 3. Without FXI, monetary policy trades off inflation and output due to the lack of risk-sharing. As described in Figure 3, when goods are substitutes, a positive local productivity shock generates an excess demand for local households (positive risk-sharing wedge). This increases the marginal cost of production and inflation in the local economy. To stabilize inflation, the local central bank increases the interest rate. This decreases the local demand and makes the local output gap negative (similarly, the US output gap becomes positive).

As discussed in the previous section, the effect of FXI can be understood in terms of income and substitution effects. First, the income effect implies that a dollar appreciation increases the local import price and decreases local households' income (the risk-sharing wedge becomes less positive). This decreases the marginal cost of production and inflation in the local economy.

[19]In a special case where $\tilde{Y}_{Lt} = \tilde{Y}_{Ut} = 0$ for all t, Equation (16) reduces to $f_t^* = u_t$, which is the same as the optimal FXI under small-open economy (Itskhoki and Mukhin 2023). This result holds in a special case where the trade elasticity equals one ($\phi = 1$), so that productivity shocks have no effect on the risk-sharing wedge.

[20]The proposition shows that the response of local output $(\partial \tilde{Y}_{Lt}/\partial A_t)^{FXI}$ becomes less negative with FXI, but it does not exclude the possibility where the response of the local output gap becomes even positive. In the numerical example of Figure 5, the response of local output is less negative but not positive in the case with FXI than the case without FXI.

To stabilize inflation, the local central bank reduces the interest rate and makes the local output gap less negative, as described in Figure 4. Second, the substitution effect implies that when the local import price is high, households shift their demand from the US to local goods, which makes the local output gap less negative. Similarly, FXI makes the US output gap less positive. Hence, optimal FXI mitigates the monetary policy trade-off between inflation and output.

4.1.2 Case 2: Optimal Monetary Policy and FXI

To capture the intuition and derive analytical results on shock transmissions, the previous subsection focused on the simple case where monetary policy targets zero inflation and FXI is set optimally. Next, I consider a more general case where central banks set both monetary policy and FXI optimally.[21]

Proposition 2 (Optimal Monetary Policy and FXI)**.** *Assume PCP, cooperation, and commitment, and consider the case where both monetary policy and FXI are set optimally. The optimal monetary policy rules are characterized as:*

$$0 = \theta\pi_{Lt} + (\tilde{Y}_{Lt} - \tilde{Y}_{Lt-1}) + \xi_{\mathcal{W}}(\tilde{\mathcal{W}}_t - \tilde{\mathcal{W}}_{t-1}), \tag{20}$$

$$0 = \theta\pi^*_{Ut} + (\tilde{Y}_{Ut} - \tilde{Y}_{Ut-1}) - \xi_{\mathcal{W}}(\tilde{\mathcal{W}}_t - \tilde{\mathcal{W}}_{t-1}), \tag{21}$$

and the optimal FXI rule is characterized as:

$$f^*_t = -\xi_Y\{(\tilde{Y}_{Lt} - E_t\tilde{Y}_{Lt+1}) - (\tilde{Y}_{Ut} - E_t\tilde{Y}_{Ut+1})\} + u_t, \tag{22}$$

where $\xi_{\mathcal{W}}, \xi_Y > 0$ are constants given in the appendix.

Proof. See Appendix A.4.

The optimal monetary policy for the local central bank is to set the nominal interest rate so that Equation (20) is implicitly satisfied. First, as in the closed-economy model, the local interest should increase (which lowers inflation) when the growth rate of the output gap is positive. Second, when the growth rate of the risk-sharing wedge is positive, local households

[21] Corsetti et al. (2023) studies the optimal monetary policy rule when FXI is not available. See Appendix A.4 for derivation.

have excess demand over US households so the interest rate should increase. The optimal monetary policy for the US central bank is characterized similarly as Equation (21). The optimal FXI can be characterized similarly to the case where monetary policy targets inflation.

Proposition 2 characterizes a general case where central banks optimize both monetary policy and FXI. However, the limitation of this approach is that it is difficult to analytically characterize the shock transmission mechanism, unlike the previous simple case where monetary policy follows inflation targeting. In the next subsection, I calibrate the model and show quantitatively that FXI has an economically meaningful effect on inflation-output stabilization.

4.2 Quantitative Results

In this section, I conduct numerical exercises to validate the analytical findings in the previous section and quantify the effect of FXI. I first calibrate the model to match the observed data on FXI and UIP deviation for major currencies. Next, I show that the optimal FXI rule discussed in the previous section has a quantitatively meaningful effect on stabilizing inflation and output.

Parameter Values. The calibration mostly adopts conventional values in the international business cycle literature, as summarized in Table 1.[22] I calibrate the model at the quarterly frequency. I set $\beta = 0.995$ to target the annual interest rate of 2%. I set the risk aversion to $\sigma = 2$, which is a standard value in the international macroeconomics literature (Itskhoki and Mukhin 2021). I set the inverse Frisch elasticity of labor to $\eta = 0.35$ (Bodenstein et al. 2023). I set the trade openness to $a = 0.88$ to match the trade-to-GDP ratio in large open economies, such as Japan and the United States (Itskhoki and Mukhin 2021). I choose the labor disutility parameter of ζ_l to match the steady-state labor $\bar{L} = 0.33$. There is a wide range of estimates for the trade elasticity ϕ. Estimates using microdata are as high as 4 (Bernard et al. 2003), while values lower than 1 can also be empirically relevant (Corsetti et al. 2008). In the benchmark exercise, I set $\phi = 2$, a standard value in the international business cycle literature (Bodenstein et al. 2023). I set the elasticity of substitution between differentiated goods to $\theta = 10$ so that the price markup is 11%. I set the Calvo price stickiness to $\xi_p = 0.75$ to target the average duration

[22]To provide a sharp analytical characterization of optimal policy rules, the calibration does not target empirical moments in particular countries but follows broader literature on international macroeconomics. A more general setup would allow for asymmetries across countries, including trade openness and country size, helping the model to match empirical moments.

Table 1: Parameter Values

Value	Description	Notes
$\beta = 0.995$	Discount factor (local)	Annual interest rate = 2%
$\sigma = 2$	Relative risk aversion	Itskhoki and Mukhin (2021)
$\eta = 0.35$	Inverse Frisch elasticity	Bodenstein et al. (2023)
$\zeta_l = 13.3$	Labor disutility (local)	Steady-state labor = $1/3$
$a = 0.88$	Home bias of consumption	Bodenstein et al. (2023)
$\phi = 2.0$	CES Local & US goods	Bodenstein et al. (2023)
$\theta = 10$	CES differentiated goods	Price markup = 11%
$\xi_p = 0.75$	Calvo price stickiness	Duration of four quarters
$\bar{\pi} = 1$	Steady-state inflation	
$\omega = 1.53$	Elasticity of UIP deviation to FXI	$\Delta(UIP)/\Delta(FXI/GDP) = 0.51$
$\rho_a = 0.97$	Persistence of productivity shock	Itskhoki and Mukhin (2021)
$\rho_\mu = 0.2$	Persistence of markup shock	Bodenstein et al. (2023)
$\sigma_a = 0.015$	SD of productivity shock	Bodenstein et al. (2023)
$\sigma_\mu = 0.011$	SD of markup shock	Bodenstein et al. (2019)

Note: The table shows the parameter settings for impulse response analyses in subsection 4.2.

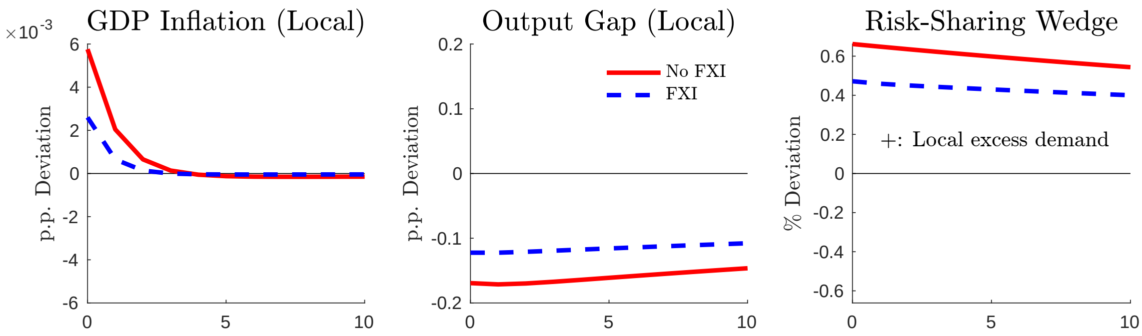
of price stickiness of four quarters. I set $\omega = 1.42$ to match the observed data on FXI and UIP deviation for major currencies (see Appendix A.5 for details). Finally, I set the persistence of productivity shocks to $\rho_a = 0.97$ following Itskhoki and Mukhin (2021) and the persistence of markup shocks to $\rho_\mu = 0.2$ following Bodenstein et al. (2023).

I quantify the effect of FXI based on this calibration. I consider two sources of shocks, productivity and markup, and compare impulse responses when central banks do not intervene and when they intervene cooperatively. This section focuses on the case where central banks set both monetary policy and FXI optimally.[23]

Productivity Shocks. Figure 5 shows impulse responses to a one-standard-deviation increase in the local productivity. To understand the mechanism, I focus on the responses of three variables: local producer-price (GDP) inflation rate, the local output gap in the local economy, and the risk-sharing wedge across countries (Figure A1 provides impulse responses of other variables). The red solid line shows the case where monetary policy is set optimally but FXI is not available (Corsetti et al. 2023) and the blue dashed line shows the case where both monetary

[23] Figure A1 provides the full impulse response results, comparing the case where central banks (i) optimize FXI under inflation-targeting monetary policy and (ii) optimize both monetary policy and FXI.

Figure 5: Impulse Responses to a Local Productivity Increase



Note: This figure plots impulse responses to a one-standard-deviation increase in local productivity. The red and blue lines show the results under no intervention and cooperative intervention cases, respectively.

policy and FXI are set optimally (Proposition 1).

As discussed in Section 3, when FXI is not available (the red line), an increase in local productivity generates local excess demand and makes the risk-sharing wedge positive. This increases the marginal cost of production and inflation in the local economy. To stabilize inflation, central banks increase the monetary policy rate, making the output gap negative.

However, when both monetary policy and FXI are available (the blue line), the optimal policy is to buy dollars and sell the local currency using FXI. A dollar appreciation increases the import prices faced by local households, which reduces the risk-sharing wedge and generates downward pressure on inflation (income effect). At the same time, households shift their demand from US to local goods, making the local output gap less negative (substitution effect).[24]

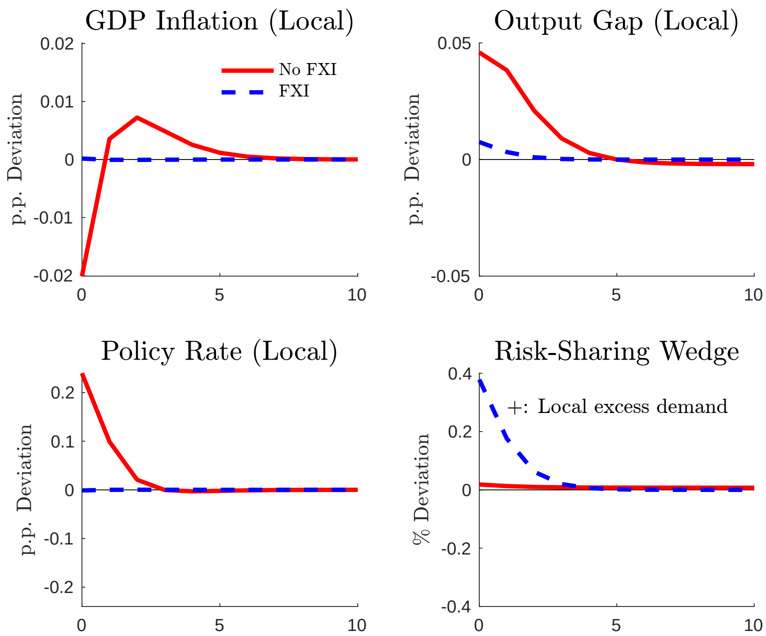
Cost-Push Shocks. I have focused on the productivity shock to capture intuition. However, in reality, inflation can also be driven by cost-push shocks, such as the pandemic and war. In particular, I consider an inflationary cost-push (markup) shock in the US and its transmission to the local economy. This provides the most interesting implication on how FXI can insulate economies from international spillovers of shocks.

Figure 6 plots the impulse responses of the local inflation rate, output gap, and policy rate to a one-standard-deviation increase in the US markup.[25] When FXI is not available (the red line),

[24]The optimal FXI does not perfectly close the output gap and risk-sharing wedge. This is due to the intertemporal substitution effect of FXI. As shown in Equation (15), a purchase of US dollars appreciates the US dollar and increases the relative return on the local currency over the dollar. Hence, FXI reduces the local excess demand (narrows the risk-sharing wedge) today and increases the local excess demand (widens the risk-sharing wedge) in the future. In this sense, optimal FXI smooths the risk-sharing wedge over time.

[25]See Figure A2 for the full impulse responses.

Figure 6: Impulse Responses to a US Cost-Push Inflation



Note: This figure plots impulse responses to a one-standard-deviation increase in the US markup. The red and blue lines show the results under no intervention and cooperative intervention cases, respectively.

the local central bank cannot stabilize domestic inflation and output at the same time. As shown in the red line, US cost-push inflation appreciates the dollar and decreases the local demand. This terms-of-trade effect decreases the marginal cost of production and inflation. At the same time, since US demand for local goods increases, the local output gap becomes positive. The local central bank increases the policy rate to stabilize the output gap.[26]

However, when FXI is available (the blue line), the local central bank stabilizes both inflation and the output gap with little changes in the policy rate. The optimal FXI is to buy the local currency and sell dollars so that the local currency appreciates against the dollar. On the one hand, local households face lower import prices, which generates excess demand for local households (positive risk-sharing wedge). This causes inflationary pressure and stabilizes inflation (income effect). On the other hand, since US households reduce demand for local goods, the local output gap decreases (substitution effect). This allows central banks to stabilize inflation and output with little changes in the monetary policy rate at the cost of distorting risk-sharing.

[26]The local interest rate is implicitly determined to satisfy the optimal monetary policy rule in Equation (20). On the one hand, lower inflation generates downward pressure on the policy rate, On the other hand, higher output generates upward pressure on the policy rate. Whether a US cost-push shock increases or decreases the local policy rate depends on the relative weights of inflation and output stabilization assigned by central banks.

The key takeaway is that without FXI, external shocks weaken monetary policy independence, making it difficult for central banks to stabilize domestic inflation and output at the same time. However, by optimally combining the monetary policy and FXI, central banks can stabilize inflation with little changes in the monetary policy rate. In this sense, FXI improves monetary independence and complements the monetary policy, while at the cost of distorting resource allocation across countries.

A consensus in the small-open economy literature (Cavallino 2019, Basu et al. 2020, Itskhoki and Mukhin 2023) is that monetary policy and FXI are two separate policy tools with distinct objectives: monetary policy stabilizes inflation while FXI stabilizes capital flow (UIP) shocks. My large-open economy framework shows that the two policies have interrelated objectives: by affecting the exchange rate, FXI affects the global demand for goods in different countries and currencies, which mitigates the monetary policy trade-off between inflation and output. Optimal FXI faces a non-trivial trade-off between stabilizing internal objectives (inflation and output) and external objectives (UIP deviation and risk-sharing).

5 Case Study: Dollar Dominance in International Trade

The previous sections have shown that FXI complements monetary policy by allowing monetary authorities to stabilize inflation and output with small changes in the monetary policy rate. This section builds on this framework and studies an important and related question: what is the role of FXI in a dollarized world? This section finds a novel relationship between the dollar dominance in international trade and capital flow management in international finance. These two strands of literature are often discussed in two separate contexts. This section bridges the gap in the literature and shows that FXI is powerful under dollar dominance.[27]

In the previous sections, for analytical tractability, I have focused on a simple PCP case where export prices are sticky in the exporters' currency. While PCP assumption provides the simplest analytical solution, data suggests that exports and imports are mainly invoiced in US dollars (Gopinath et al. 2020). I introduce an alternative assumption of dominant currency pricing (DCP) or dollar pricing where both export and import prices are sticky in dollars.

This DCP assumption has two major implications. First, identical local goods are priced

[27]Another important research agenda is the role of dollars in the international financial market. Rodnyansky et al. (2024) empirically finds that FXI mitigates international spillovers of US monetary policy shocks via corporate balance sheets for firms with dollar-denominated debt.

differently in two currencies, i.e., the LOOP does not hold for local goods. When the dollar appreciates, local goods are sold at higher prices in the US than in the local economy and vice versa.[28] As discussed in Engel (2011), if firms face the same marginal cost of production irrespective of countries where goods are sold, it is inefficient for firms to sell goods at different prices. Under cooperation, central banks target this cross-currency price dispersion across countries. Second, exchange rates have asymmetric pass-through to import prices across countries. Since both export and import prices are sticky in dollars, exchange rate movements directly affect consumer prices in the local economy but have muted effects on consumer prices in the US.

5.1 Model Environment under DCP

I first describe the difference in model environments under DCP and PCP cases. Under DCP, the price of local goods is sticky in local currency in the local economy ($P_t(l)$) and sticky in the dollars in the US ($P_t^*(l)$). The local firms' maximization problem is:

$$\max_{\{P_t(l), P_t^*(l)\}} E_t \left\{ \sum_{k=0}^{\infty} Q_{t,t+k} \xi_p^k \begin{pmatrix} (1+s_t)\left[P_t(l) Y_{t+k}(l) + \mathcal{E}_t P_t^*(l) Y_{t+k}^*(l)\right] \\ -MC_{t+k}(l)\left[Y_{t+k}(l) + Y_{t+k}^*(l)\right] \end{pmatrix} \right\}.$$

Solving the maximization problem, the optimal prices of local goods charged to local and US households are:

$$\begin{aligned} P_t(l) &= \frac{\theta}{(\theta-1)(1+s_t)} \frac{E_t \sum_{k=0}^{\infty} Q_{t,t+k} \xi_p^k Y_{t+k}(l) MC_{t+k}}{E_t \sum_{k=0}^{\infty} Q_{t,t+k} \xi_p^k Y_{t+k}(l)}, \quad \text{and} \\ P_t^*(l) &= \frac{\theta}{(\theta-1)(1+s_t)} \frac{E_t \sum_{k=0}^{\infty} Q_{t,t+k} \xi_p^k Y_{t+k}^*(l) MC_{t+k}}{E_t \sum_{k=0}^{\infty} Q_{t,t+k} \xi_p^k \mathcal{E}_{t+k} Y_{t+k}^*(l)}. \end{aligned}$$

The laws of motion of local good prices faced by local and US households can be written as:

$$\begin{aligned} P_{Lt}^{1-\theta} &= \xi_p P_{Lt-1}^{1-\theta} + (1-\xi_p) P_t(l)^{1-\theta}, \quad \text{and} \\ P_{Lt}^{*1-\theta} &= \xi_p P_{Lt-1}^{*1-\theta} + (1-\xi_p) P_t^*(l)^{1-\theta}. \end{aligned}$$

[28]For example, suppose that a Japanese good's price is sticky at 100 Japanese yen (JPY) in Japan and 1 US dollar (USD) in the US. When the exchange rate is 1 USD = 100 JPY, the law of one price holds. However, when the dollar appreciates and becomes 1 USD = 110 JPY, the good is sold at 100 JPY in Japan and 110 JPY in the US when converted to the same JPY. Hence, the law of one price does not hold.

I define $\Delta_{Lt} \equiv \mathcal{E}_t P^*_{Lt}/P_{Lt}$ as the deviation from the LOOP for the local goods. The numerator and the denominator are the prices of local goods charged to local and US households, respectively, expressed in the local currency. Under PCP, the LOOP holds so $\Delta_{Lt} = 1$. However, under DCP, the LOOP does not hold so $\Delta_{Lt} \neq 1$. When P_{Lt} and P^*_{Lt} are sticky, a dollar appreciation (an increase in $\mathcal{E}_t$) increases Δ_{Lt}, so US households are charged higher prices for the identical local goods than local households despite the same marginal cost of production.

Solving the maximization problem, the NKPCs for local goods consumed by local and US households can be written as:

$$\pi_{Lt} = \beta\pi_{Lt+1} + \kappa\{(\sigma+\eta)\tilde{Y}_{Lt} - (1-a)[2a(\sigma\phi-1)(\tilde{\mathcal{T}}_t + \tilde{\Delta}_{Lt}) + (\tilde{\mathcal{W}}_t + \tilde{\Delta}_{Lt})] + \mu_t\}, \tag{23}$$

$$\pi^*_{Lt} = \beta\pi^*_{Lt+1} + \kappa\{(\sigma+\eta)\tilde{Y}_{Lt} - (1-a)[2a(\sigma\phi-1)(\tilde{\mathcal{T}}_t + \tilde{\Delta}_{Lt}) + (\tilde{\mathcal{W}}_t + \tilde{\Delta}_{Lt})] - \tilde{\Delta}_{Lt} + \mu_t\}, \tag{24}$$

and the NKPC for US goods is the same as Equation (9). We learn that π_{Lt} and π^*_{Lt} are decreasing functions of Δ_{Lt}. Intuitively, when Δ_{Lt} is high, local households face lower prices for local goods than US households. This increases local consumption demand and the marginal cost of production for local firms, leading to high inflation for locally produced goods.

5.2 Optimal Policy and Shock Transmission under DCP

Based on this environment, I derive the optimal FXI rule under DCP analytically. Next, I conduct a quantitative exercise and contrast the results with the PCP case.

As shown in Corsetti et al. (2020), under DCP, central banks' maximization problem is equivalent to minimizing the following loss function:

$$\mathcal{L} = -E_0 \sum_{t=0}^{\infty} \beta^t \frac{1}{2} \begin{bmatrix} (\sigma+\eta)\left(\tilde{Y}^2_{Lt} + \tilde{Y}^2_{Ut}\right) + \frac{\theta}{\kappa}\left(a\pi^2_{Lt} + (1-a)\pi^{*2}_{Lt} + \pi^{*2}_{Ut}\right) \\ -\frac{2a(1-a)(\sigma\phi-1)\sigma}{4a(1-a)(\sigma\phi-1)+1}\left(\tilde{Y}_{Lt} - \tilde{Y}_{Ut}\right)^2 \\ +\frac{2a(1-a)\phi}{4a(1-a)(\sigma\phi-1)+1}\left(\tilde{\mathcal{W}}_t + \tilde{\Delta}_{Lt}\right)^2 \end{bmatrix}. \tag{25}$$

There are two key differences compared to the PCP case (Equation (13)). First, the central banks take into account the weighted sum of the inflation rates of local goods consumed by local and US households (π_{Lt}, π^*_{Lt}) and the inflation rate of US goods consumed by US households π^*_{Ut}. Second, the loss depends on the deviation from the LOOP ($\tilde{\Delta}_{Lt}$).

The following proposition characterizes the optimal FXI rule under DCP analytically. Under DCP, an analytical derivation of optimal policy rules is difficult when central banks optimize both monetary policy and FXI. Hence, I derive the optimal FXI rule under inflation targeting and compare the result with the PCP case in Proposition 1.

Proposition 3 (Optimal FXI under Dollar Pricing)**.** *Assume DCP, cooperation, and commitment, and consider the case where monetary policy targets* $\pi_{Lt} = \pi^*_{Ut} = 0$ *for all* t*. The optimal FXI rule is characterized as:*

$$f^*_t = -\xi_Y^{DCP}\{(\tilde{Y}_{Lt} - E_t\tilde{Y}_{Lt+1}) - (\tilde{Y}_{Ut} - E_t\tilde{Y}_{Ut+1})\} - \xi_\Delta^{DCP}(\tilde{\Delta}_{Lt} - E_t\tilde{\Delta}_{Lt+1}) + u_t, \tag{26}$$

where ξ_Y^{DCP}, $\xi_W^{DCP} > 0$ *are constants given in the appendix.*

Proof. See Appendix A.6.

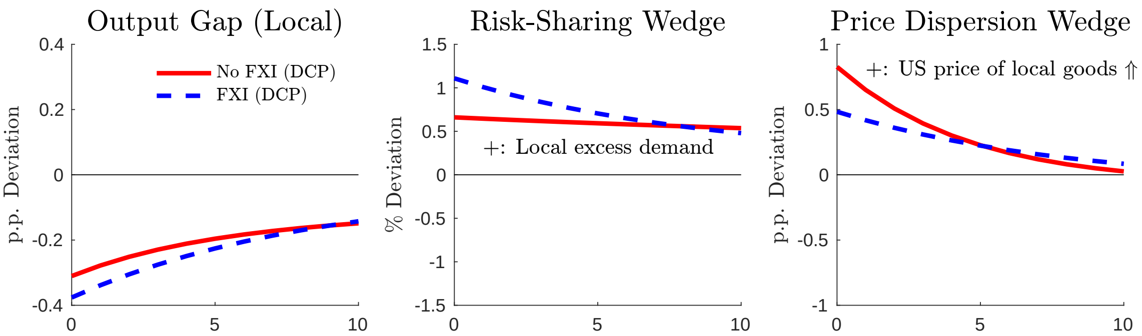
The key difference between the DCP and PCP cases is that, under DCP, the optimal FXI responds to the cross-currency price dispersion wedge. When the dollar price of local goods is high ($\tilde{\Delta}_{Lt} > 0$), the optimal FXI is to buy the local currency and sell dollars ($f^*_t < 0$). Intuitively, when the dollar appreciates, identical local goods are priced higher in the US than the local economy when converted into the same currency. The optimal FXI is to buy the local currency and sell dollars, which depreciates dollars and mitigates the price-dispersion gap. Hence, incomplete exchange-rate pass-through generates an additional trade-off of FXI between internal objectives (inflation and output gap in each country) and external objectives (risk-sharing and price dispersion across countries).

To quantify the importance of the price-dispersion wedge, Figure 7 plots the impulse responses of the local output gap, risk-sharing wedge, and price-dispersion wedge to a one-standard-deviation increase in local productivity.[29] Similarly to the PCP case (Figure 5), when goods are substitutes, an increase in local productivity generates local excess demand, causing upward pressure on inflation. To stabilize inflation, the local central bank increases the interest rate, which in turn makes the output gap negative. The key difference in the DCP case is the price dispersion wedge: an increase in local productivity depreciates the local currency against the dollar and increases the price of local goods sold in the US (positive $\tilde{\Delta}_{Lt}$).

The optimal FXI faces a trade-off between stabilizing internal objectives (inflation and

[29] Figure A3 shows impulse responses of all variables to productivity and cost-push shocks.

Figure 7: Impulse Responses to a Local Productivity Increase under DCP



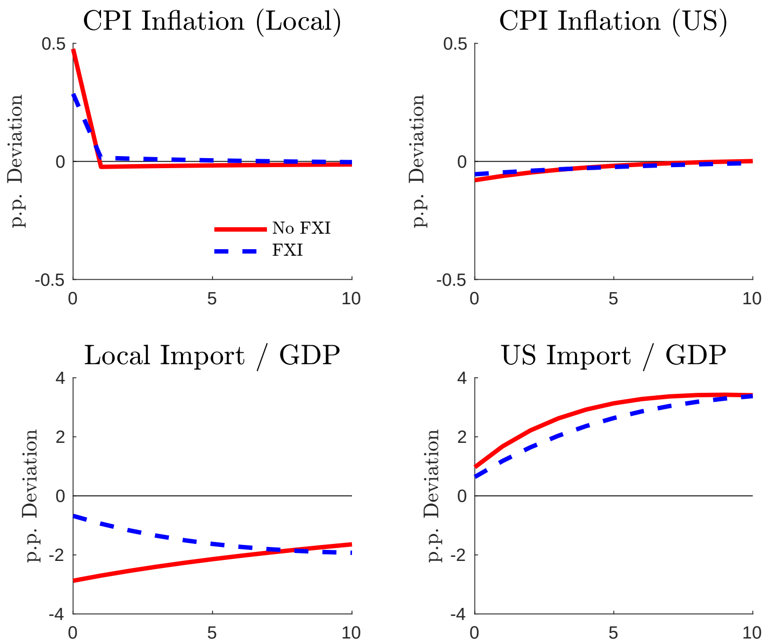
Note: This figure plots impulse responses to a one-standard-deviation increase in local productivity under DCP. The red and blue lines show the results under no intervention and cooperative intervention cases, respectively.

output) and external objectives (risk-sharing and price dispersion). As in the PCP case, US dollar purchases stabilize the output gap and risk-sharing wedge. However, US dollar sales stabilize the price-dispersion wedge (decreases $\tilde{\Delta}_{Lt}$) by appreciating the local currency and reducing the price of local goods sold in the US. My quantitative exercise shows that the optimal policy under DCP is to sell dollars and purchase the local currency, which widens the output gap and risk-sharing wedge but narrows down the price-dispersion wedge. This is in contrast to the PCP case, where the optimal FXI is to purchase dollars. The key insight is that the direction of the optimal FXI depends on the exchange rate pass-through.

To further understand the mechanism, I show that FXI has asymmetric transmission mechanisms across the two countries under DCP. Figure 8 plots the impulse responses of local and US CPI inflation rates (top panels) and import over GDP ratio (bottom panels) to a one-standard-deviation increase in local productivity. We learn that FXI is particularly effective in stabilizing local CPI inflation, while the effect on US CPI inflation is small. Since local import prices are sticky in dollars, an appreciation of the local currency directly reduces import prices. This generates large expenditure-switching from local to US goods and increases imports via the substitution effect. At the same time, lower import prices increase households' income and generate excess demand (positive risk-sharing wedge) via the income effect. In contrast, since US import prices are denominated in their domestic currency, the US dollar, an appreciation of dollars does not increase CPI inflation and dampens imports by a large degree. In other words, the income and substitution effects of FXI are muted for US households.

This analysis shows that under DCP, FXI is powerful in stabilizing local inflation and consumption demand, with limited spillover effects on US inflation and demand. This helps

Figure 8: Asymmetric Expenditure Switching Channel under DCP



Note: This figure plots impulse responses to a one-standard-deviation increase in local productivity under DCP. The red and blue lines show the results under no intervention and cooperative intervention cases, respectively.

explain why FXI is widely used in a modern dollarized world.

6 Robustness Checks

This section provides robustness checks to support the main results. subsection 6.1 studies non-cooperative FXI against US tariff shocks, focusing on an analytically tractable case. subsection 6.2 studies the transmission channel of preference shocks.

6.1 Non-Cooperative FXI against US Tariff Shocks: An Analytically Tractable Case

The previous sections assumed that monetary policy and FXI are set cooperatively to maximize the sum of the welfare of representative households in the two countries. However, in reality, central banks may target domestic inflation and output, ignoring international spillovers. For example, countries such as China and Japan are on the US monitoring list as they accumulate trillions of dollars in foreign exchange reserves to devalue their currencies and increase exports. On the other hand, US President Donald Trump imposes tariffs to limit imports and protect domestic industry. To check the robustness of my analysis, this section deviates from the assumption of international cooperation and studies nationally oriented interventions. As a policy-relevant case study, I consider an exogenous increase in the US tariff on imports and study the effect of nationally oriented FXI by the local central bank.

Modern macroeconomics literature faces many challenges characterizing full strategic interaction of monetary policy in dynamic games. Instead, I focus on the analytically tractable case, assuming that central banks in each country only target domestic inflation and output gap but do not take into account international objectives, including the exchange rate misalignment and risk-sharing wedge. This flexible inflation targeting is a convention followed by many central banks (Svensson 2003; 2010) and called a "keep-your-house-in-order" policy (Bodenstein et al. 2023).

I make the following assumptions to focus on the analytically tractable case. First, I consider a situation where the local central bank uses both monetary policy and FXI, but the US central bank (the Fed) only uses monetary policy. This is because, historically, the US has intervened infrequently in the foreign exchange market. Second, following Svensson (2003; 2010), the local central bank minimizes the weighted sum of the domestic inflation rate and output gap:

$$\mathcal{L} = -E_0 \sum_{t=0}^{\infty} \beta^t \frac{1}{2} \left[\frac{\theta}{\kappa} \pi_{Lt}^2 + (\sigma + \eta) \tilde{Y}_{Lt}^2 \right]. \tag{27}$$

To further simplify the problem, I assume that the monetary policy in each country strictly targets domestic inflation ($\pi_{Lt} = \pi_{Ut}^* = 0$). Conditional on inflation-targeting monetary policy, the optimal FXI by the local central bank can be characterized as a simple rule targeting the expected growth rate of the local output gap.

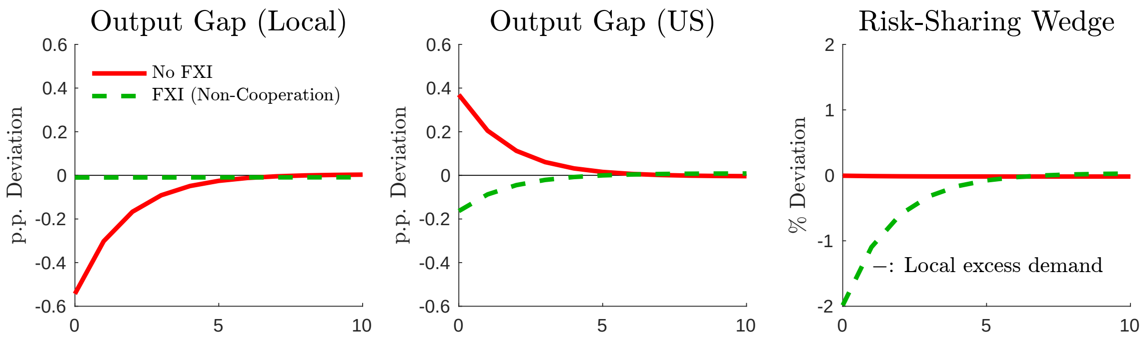
Proposition 4 (Optimal Non-Cooperative FXI: An Analytically Tractable Case)**.** *Assume PCP, non-cooperation, and commitment, and consider the case where monetary policy targets $\pi_{Lt} = \pi_{Ut}^* = 0$ for all t. The optimal FXI rule is characterized as:*

$$E_t \tilde{Y}_{Lt+1} - \tilde{Y}_{Lt} = 0. \tag{28}$$

Proof. See Appendix A.7.

Let τ_t^* be the tariff on imported goods consumed by US households. I assume that the tariff

Figure 9: Impulse Responses to a US Tariff Increase



Note: This figure plots impulse responses to a one-standard-deviation increase in US tariff. The red and green lines show the results under no intervention and non-cooperative intervention cases, respectively.

follows an exogenous AR(1) process with shocks and that the tariff revenue is transferred to US households in a lump-sum way:

$$\tau_t^* P_{Lt}^* C_{Lt}^* = T_t^*.$$

I conduct a numerical exercise to quantify the effect of FXI. Following Barattieri et al. (2021) and Bergin and Corsetti (2023), I set the persistence and standard deviation of tariff shock at $\rho_\tau = 0.56$ and $\sigma_\tau = 0.034$. Figure 9 plots impulse responses to a one-standard-deviation increase in the US tariff. The red and green lines show the results under no intervention and non-cooperative intervention cases, respectively. Without FXI, the US tariff shock makes the local output gap negative, the US output gap positive, and the risk-sharing wedge negative, although the effect on the risk-sharing wedge is quantitatively small. However, the optimal FXI achieves nearly full stabilization of the local output gap but reverses the US output gap from positive to negative and makes the risk-sharing gap significantly negative.

Intuitively, without FXI, a US tariff increases the import prices faced by US households. Since US households shift their demand from local to US goods, the local output gap becomes negative, and the US output gap becomes positive. At the same time, US tariffs increase consumer price inflation and appreciate the dollar in real terms under the home bias of consumption. Hence, local households face higher import prices and lower income, making the risk-sharing wedge negative.

The local central bank responds by buying dollars using FXI. The effect of FXI can be understood in terms of income and substitution effects as discussed in Section 4. First, the income effect implies that an appreciation of the dollar lowers local households' income due to higher import prices (negative risk-sharing wedge), which lowers the marginal cost of production and inflation. To stabilize inflation, the local central bank lowers the interest rate and increases output. Second, the substitution effect implies that households shift their demand from US to local goods so that the local firms increase their exports to US households. As a result, FXI makes the local output gap less negative and the US output gap less positive. However, the trade-off is that FXI further widens the negative risk-sharing wedge, disproportionately increasing US households' consumption demand than local households' demand.

My model provides a rationale for why we need to convince policymakers to coordinate on exchange rate stabilization. Monetary and exchange rate policies exclusively focusing on domestic inflation and output stabilization not only generate an inflation-output trade-off for foreign policymakers (beggar-thy-neighbor) but also distort the world resource allocation in favor of foreign households over domestic households (beggar-thy-self).

6.2 Preference Shocks

While previous sections focused on supply-side shocks, including productivity and markup, this section studies demand-side shocks, especially shocks to households' preferences.

I assume that the households' lifetime utility has the following form:

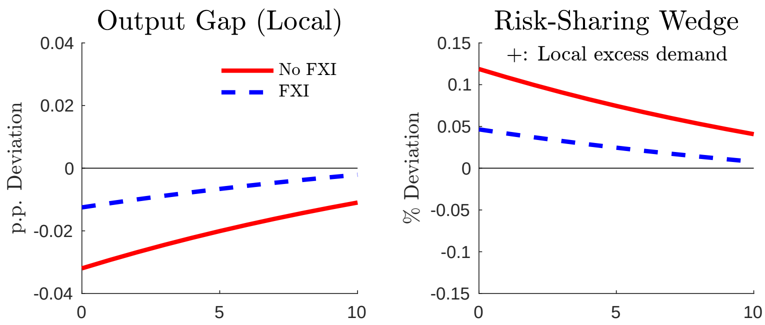
$$E_0 \sum_{t=0}^{\infty} \beta^t \iota_t \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \zeta_l \frac{L_t^{1+\eta}}{1+\eta} \right],$$

where ι_t is an exogenous intertemporal preference shock that follows an AR(1) process. An increase in ι_t implies that households put a high weight on period-t consumption, so they increase consumption today and decrease consumption in the future. Following Corsetti et al. (2010), I assume that the mean and variance of preference shock at $\rho_\iota = 0.95$ and $\sigma_\iota = 0.001$. I assume that the central banks in the two countries strictly target zero producer-price inflation.

Figure 10 plots impulse response to a one-standard-deviation positive preference shock.[30] The red and blue lines show the results under no intervention and cooperative intervention cases, respectively. Without FXI, when ι_t increases, the marginal utility of consumption for local households increases, generating local excess demand (positive risk-sharing wedge). This local excess demand increases the marginal cost of production and generates inflationary pressure. To stabilize inflation, the local central bank increases the monetary policy rate, which

[30] See Figure A5 for the full impulse responses.

Figure 10: Impulse Responses to a Positive Local Preference Shock



Note: This figure plots impulse responses to a one-standard-deviation positive shock to the local households' preference. The red and blue lines show the results under no intervention and cooperative intervention cases, respectively.

reduces the local output. The optimal FXI is to buy dollars and sell the local currency, which appreciates the dollar and closes the output gap and risk-sharing wedge. The intuition is similar to the case with a positive productivity shock (Figure 5).

7 Conclusion

Conventional wisdom in the literature is that monetary policy and FXI are two distinct policy tools with separate objectives: optimal monetary policy exclusively targets inflation, while optimal FXI independently stabilizes the exchange rate. This consensus is based on a small open economy model where agents take the international price as given. However, this consensus changes drastically in large open economies where FXI can affect international prices and demand. Should policies respond to domestic inflation or global business cycle conditions and imbalances?

In this paper, I have shown that FXI in large open economies faces a non-trivial trade-off between internal and external objectives. On the one hand, FXI mitigates the inflation-output trade-off of monetary policy by stabilizing domestic inflation and demand without large changes in policy interest rates. On the other hand, FXI distorts world inflation and demand since FXI affects international prices. This trade-off between internal and external objectives implies that monetary policy and FXI are not completely independent policy instruments, but a policy encompassing both instruments achieves the optimal outcome.

Furthermore, I find that FXI is powerful in stabilizing domestic inflation when goods traded in international markets are priced in US dollars. This result explains the popularity of FXI in a dollarized world. Finally, FXI that exclusively targets domestic objectives exacerbates foreign inflation-output trade-off and resource misallocation across countries.

There are several important and challenging directions for future research. The first is to develop an identification method of FXI and empirically estimate its effects. Fratzscher et al. (2019) compares different identification methods, including reaction function, propensity score matching, instrumental variables, and high-frequency approaches. Recent works exploit granular firm- or bank-level data and combine their balance sheet information with credit supply and employment (Gonzalez et al. 2021) or with stock prices (Rodnyansky et al. 2024). Another policy question is how to combine FXI with other capital account management policies, including capital control and macroprudential policy, in the context of IMF's integrated policy framework (Basu et al. 2020). Finally, an important research area is the political economy of unconventional policy tools and the possibility that they are abused by policymakers, as discussed in Maggiori (2022). Future research is needed to characterize the strategic interaction of FXI between policymakers in dynamic games and study the political costs of nationally oriented monetary and exchange rate policies.

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Appendix

A Derivations and Proofs

A.1 Useful Equilibrium Relationships

This section provides equilibrium first-order relationships which are useful for proofs of propositions in Section 2. The derivation follows Corsetti et al. (2010; 2023).

I focus on the LCP case (the PCP case can be derived analogously by setting $\Delta_t = 0$). Let the variables with hat denote the deviation from the steady state. For simplicity, assume symmetry so that $\mathcal{E}_t P^*_{Lt}/P_{Lt} = \mathcal{E}_t P^*_{Ut}/P_{Ut} = \Delta_t$. By the definition of real exchange rate, it is expressed in terms of terms of trade and price dispersion:

$$e_t = \frac{\mathcal{E}_t P^*_t}{P_t} = \frac{\mathcal{E}_t \left[a(P^*_{Lt})^{1-\phi} + (1-a)(P^*_{Ut})^{1-\phi}\right]^{\frac{1}{1-\phi}}}{\left[a(P_{Lt})^{1-\phi} + (1-a)(P_{Ut})^{1-\phi}\right]^{\frac{1}{1-\phi}}}$$

$$= \frac{\left[a\left(\frac{\mathcal{E}_t P^*_{Lt}}{P_{Lt}}\right)^{1-\phi} + (1-a)\left(\frac{\mathcal{E}_t P^*_{Ut}}{P_{Lt}}\right)^{1-\phi}\right]^{\frac{1}{1-\phi}}}{\left[a + (1-a)\left(\frac{P_{Ut}}{P_{Lt}}\right)^{1-\phi}\right]^{\frac{1}{1-\phi}}}, \tag{A1}$$

where:

$$\frac{\mathcal{E}_t P^*_{Ut}}{P_{Lt}} = \frac{\mathcal{E}_t P^*_{Lt}}{P_{Lt}} \frac{P_{Ut}}{\mathcal{E}_t P^*_{Lt}} \frac{\mathcal{E}_t P^*_{Ut}}{P_{Ut}} = \Delta_t^2 \mathcal{T}_t, \tag{A2}$$

$$\frac{P_{Ut}}{P_{Lt}} = \frac{\mathcal{E}_t P^*_{Lt}}{P_{Lt}} \frac{P_{Ut}}{\mathcal{E}_t P^*_{Lt}} = \Delta_t \mathcal{T}_t. \tag{A3}$$

Log-linearizing Equation (A1), we obtain Equation (12).

Next, I approximate the aggregate demand. Under the assumption of symmetry, we have:

$$\hat{Y}_{Lt} + \hat{Y}_{Ut} = \hat{C}_t + \hat{C}^*_t = 0. \tag{A4}$$

Combining Equations (6) and (A4) gives:

$$\hat{Y}_{Lt} - \hat{C}_t = \hat{C}^*_t - \hat{Y}_{Ut} = \frac{1}{2}[\hat{Y}_{Lt} - \hat{Y}_{Ut} - \sigma^{-1}(\hat{Q}_t + \tilde{\mathcal{W}}_t)]. \tag{A5}$$

Substituting Equation (2) into the aggregate demand $Y_{Lt} = C_{Lt} + C^*_{Lt}$ for local good gives:

$$Y_{Lt} = \left(\frac{P_{Lt}}{P_t}\right)^{-\phi} [aC_t + (1-a)(e_t\Delta_t^{-1})^{\phi} C^*_t]. \tag{A6}$$

Log-linearizing the CPI (1) gives:

$$\hat{P}_t - \hat{P}_{Lt} = (1-a)(\hat{\mathcal{T}}_t + \tilde{\Delta}_t). \tag{A7}$$

Using Equation (A7), (A6) can be log-linearized as:

$$\hat{Y}_{Lt} - \hat{C}_t = (1-a)\sigma^{-1}[\sigma\phi(\hat{e}_t + \hat{\mathcal{T}}_t) - \hat{e}_t - \tilde{\mathcal{W}}_t]. \tag{A8}$$

Using Equations (12), (A8) can be rewritten as:

$$\hat{Y}_{Lt} - \hat{C}_t = (1-a)\sigma^{-1}[2a\sigma\phi(\hat{\mathcal{T}}_t + \tilde{\Delta}_t) - \hat{e}_t - \tilde{\mathcal{W}}_t]. \tag{A9}$$

Combining the two expressions (A5) and (A9) for the aggregate demand, the terms of trade can be expressed as:

$$\hat{\mathcal{T}}_t = \frac{\hat{Y}_{Lt} - \hat{Y}_{Ut} - (2a-1)\tilde{\mathcal{W}}_t}{4a(1-a)(\sigma\phi - 1) + 1}. \tag{A10}$$

A.2 Proof of Lemma 1

The proof follows the appendix of Itskhoki and Mukhin (2021). Differently from their paper, the central bank can use FXI in addition to monetary policy.

To begin with, I show the first equality of Equation (10), which describes the relationship between risk-sharing wedge and UIP deviation. The Euler equations of local and US households are characterized in log-linearized form:

$$\tilde{r}_t = \sigma E_t\left[\tilde{C}_{t+1} - \tilde{C}_t\right], \tag{A11}$$

$$\tilde{r}^*_t = \sigma E_t\left[\tilde{C}^*_{t+1} - \tilde{C}^*_t\right] \tag{A12}$$

Taking the difference of Equations (A11) and (A12) and subtracting $\Delta\tilde{e}_{t+1} = \tilde{e}_{t+1} - \tilde{e}_t$ from both sides,

$$E_t\left[\sigma\left\{(\tilde{C}_{t+1}-\tilde{C}_t)-(\tilde{C}^*_{t+1}-\tilde{C}^*_t)\right\}-\Delta\tilde{e}_{t+1}\right]=\tilde{r}_t-\tilde{r}^*_t-E_t\Delta\tilde{e}_{t+1}.$$

Using the definition of risk-sharing wedge (6), we obtain the first equality:

$$E_t\tilde{\mathcal{W}}_{t+1}-\tilde{\mathcal{W}}_t=\tilde{r}_t-\tilde{r}^*_t-E_t\Delta\tilde{e}_{t+1}. \tag{A13}$$

Next, I show the second equality of Equation (10), which describes the relationship between financial flows and UIP deviation. The maximization problem (4) of arbitrageurs can be rewritten as:

$$\max_{d_t} E_t\left\{-\frac{1}{\omega^F}\exp\left(-\omega^F\bar{R}_t(1-e^{x_t})d_t\right)\right\}, \tag{A14}$$

where $x_t=\tilde{r}^*_t-\tilde{r}_t-\Delta\tilde{e}_{t+1}$ is the nominal carry trade return. When the time period is short, x_t can be expressed as the normal diffusion process:

$$d\mathcal{X}_t=x_t dt+\sigma_e dB_t,$$

where $x_t=\tilde{r}_t-\tilde{r}^*_t-\Delta\tilde{\mathcal{E}}_{t+1}$ is the nominal carry trade return and B_t is a standard Brownian motion. Note that the excess return is equal in nominal and real terms when log-linearized:

$$\begin{aligned}x_t&=\tilde{r}_t-\tilde{r}^*_t-\Delta\tilde{\mathcal{E}}_{t+1}\\&=(\tilde{R}_t-E_t\pi_{t+1})-(\tilde{R}^*_t-E_t\pi^*_{t+1})-E_t(\Delta\tilde{\mathcal{E}}_{t+1}+\pi^*_{t+1}-\pi_{t+1})\\&=\tilde{R}_t-\tilde{R}^*_t-\Delta\tilde{e}_{t+1}.\end{aligned}$$

The maximization problem (A14) can be rewritten as:

$$\max_{D^*_t} E_t\left\{-\frac{1}{\omega^F}\exp\left(-\omega^F\bar{R}_t(1-e^{d\mathcal{X}_t})d_t\right)\right\}. \tag{A15}$$

Using Ito's lemma, the objective function can be rewritten as:

$$E_t\left\{-\frac{1}{\omega^F}\exp\left(-\omega^F\bar{R}_t\left(-d\mathcal{X}_t-\frac{1}{2}(d\mathcal{X}_t)^2\right)d_t\right)\right\}$$

$$= -\frac{1}{\omega^F}\exp\left(\left[\omega^F\left(x_t + \frac{1}{2}\sigma_e^2\right)d_t - \frac{1}{2}(\omega^F)^2\sigma_e^2 d_t^2\right]dt\right).$$

Solving the maximization problem, the optimal portfolio decision is:

$$d_t = \frac{D_t}{P_t} = m_d\frac{\tilde{R}_t - \tilde{R}_t^* - E_t\Delta\tilde{e}_{t+1} + \sigma_e^2}{\omega^F\sigma_e^2}. \tag{A16}$$

Substituting Equation (A16) and $U_t = m_u u_t$, $F_t = m_u f_t$ into the market clearing condition (5) for the local currency bond, we obtain:

$$\frac{B_t}{P_t} + \frac{1}{P_t}m_u u_t + m_d\frac{\tilde{R}_t - \tilde{R}_t^* - E_t\Delta\tilde{e}_{t+1} + \sigma_e^2}{\omega^F\sigma_e^2} + \frac{1}{P_t}m_u f_t = 0. \tag{A17}$$

Since the arbitrageurs, noise traders, and central bank (FXI) take zero net positions,

$$\frac{D_t + U_t + F_t}{R_t} = -\mathcal{E}_t\frac{D_t^* + U_t^* + F_t^*}{R_t^*}.$$

Using (5), we obtain $B_t/R_t + \mathcal{E}_t B_t^*/R_t^* = 0$. Substituting the zero net positions for households and central bank into Equation (A17) yields:

$$\frac{\tilde{R}_t - \tilde{R}_t^* - \Delta\tilde{e}_{t+1} + \sigma_e^2}{\omega\sigma_e^2/m_d} = \frac{R_t}{R_t^*}\frac{e_t}{P_t^*}f_t^* - \frac{1}{P_t}m_u u_t + \frac{R_t}{R_t^*}e_t Y_t^*\frac{B_t^*}{P_t^* Y_t^*} \tag{A18}$$

Log-linearizing this gives the second equality:

$$\tilde{r}_t - \tilde{r}_t^* - E_t\Delta\tilde{e}_{t+1} = \omega(f_t^* - u_t) + \omega_b b_t^*. \tag{A19}$$

Combining Equations (A13) and (A19), we obtain the relationship (10) between the risk-sharing wedge, UIP deviation, and net foreign asset position. □

Incomes and Losses of Carry Trade Positions. For simplicity, I assume that the profits and losses of carry-trade positions by the financiers and noise traders and interventions by the local central bank are transferred to the local households in a lump-sum way. However, the assumption on the ownership structure does not affect the first-order dynamics of the model, as discussed in Itskhoki and Mukhin (2021). To see this, combining the positions of the financiers, the noise traders, and the local central bank, the total carry trade profit can be written as:

$$\bar{R}_t(D_t + U_t + F_t) = -\bar{R}_t B_t = \bar{R}_t \bar{Y} b_t.$$

The combined carry trade profit equals the product of the UIP deviation ($\bar{R}_t$) and the households' net foreign asset position ($\bar{Y} b_t$). Each of them is first-order but their product is second-order and small enough relative to the size of the countries' budget constraint.

A.3 Proof of Proposition 1

Part (i). Under cooperation and commitment, the central banks in the two countries minimize the loss (13) subject to the NKPCs (8) and (9) and the UIP condition (15). Assuming $\pi_{Lt} = \pi^*_{Ut} = 0$ for all t, the first-order conditions can be written as:

$$\begin{aligned}
\tilde{Y}_{Lt}: \quad 0 =& -(\sigma+\eta)\tilde{Y}_{Lt} + \frac{2a(1-a)(\sigma\phi-1)\sigma}{4a(1-a)(\sigma\phi-1)+1}(\tilde{Y}_{Lt}-\tilde{Y}_{Ut}) \\
&- \frac{2a(1-a)\phi}{4a(1-a)(\sigma\phi-1)+1}\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}\tilde{\mathcal{W}}_t \\
&+ \left[\sigma+\eta-\frac{(1-a)(\sigma-1)}{2a(\phi-1)+1}\right]\kappa\gamma_{Lt} + \frac{(1-a)(\sigma-1)}{2a(\phi-1)+1}\kappa\gamma^*_{Ut} \\
&- \frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}(\lambda_t - \beta^{-1}\lambda_{t-1}),
\end{aligned} \tag{A20}$$

$$\begin{aligned}
\tilde{Y}_{Ut}: \quad 0 =& -(\sigma+\eta)\tilde{Y}_{Ut} - \frac{2a(1-a)(\sigma\phi-1)\sigma}{4a(1-a)(\sigma\phi-1)+1}(\tilde{Y}_{Lt}-\tilde{Y}_{Ut}) \\
&+ \frac{2a(1-a)\phi}{4a(1-a)(\sigma\phi-1)+1}\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}\tilde{\mathcal{W}}_t \\
&+ \left[\sigma+\eta-\frac{(1-a)(\sigma-1)}{2a(\phi-1)+1}\right]\kappa\gamma^*_{Ut} + \frac{(1-a)(\sigma-1)}{2a(\phi-1)+1}\kappa\gamma_{Lt} \\
&+ \frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}(\lambda_t - \beta^{-1}\lambda_{t-1}),
\end{aligned} \tag{A21}$$

$$\begin{aligned}
\tilde{\mathcal{B}}_t: \quad 0 =& -\frac{2a(1-a)\phi}{4a(1-a)(\sigma\phi-1)+1}(E_t\tilde{\mathcal{W}}_{t+1} - \tilde{\mathcal{W}}_t) \\
&+ (1-a)\frac{2a(\sigma\phi-1)+1}{4a(1-a)(\sigma\phi-1)+1}\kappa[(E_t\gamma_{Lt+1}-\gamma_{Lt}) - (E_t\gamma^*_{Ut+1}-\gamma^*_{Ut})] \\
&- [(E_t\lambda_{t+1} - \beta^{-1}\lambda_t) - (\lambda_t - \beta^{-1}\lambda_{t-1})],
\end{aligned} \tag{A22}$$

$$f_t: \quad \lambda_t = 0. \tag{A23}$$

Taking the difference between Equations (A20) and (A21),

$$
\begin{aligned}
0 = & \left[\sigma + \eta - \frac{4a(1-a)(\sigma\phi - 1)\sigma}{4a(1-a)(\sigma\phi-1)+1}\right](\tilde{Y}_{Lt} - \tilde{Y}_{Ut}) \\
& + \frac{4a(1-a)\phi}{4a(1-a)(\sigma\phi-1)+1}\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}\tilde{\mathcal{W}}_t \\
& - \left[\sigma + \eta - \frac{2(1-a)(\sigma-1)}{2a(\phi-1)+1}\right]\kappa(\gamma_{Lt} - \gamma^*_{Ut}).
\end{aligned} \tag{A24}
$$

I define δ_1, δ_2, and δ_3 as:

$$
\delta_1 \equiv \frac{\left[\sigma + \eta - \frac{4a(1-a)(\sigma\phi-1)\sigma}{4a(1-a)(\sigma\phi-1)+1}\right]}{\left[\sigma + \eta - \frac{2(1-a)(\sigma-1)}{2a(\phi-1)+1}\right]},
$$

$$
\delta_2 \equiv \frac{\left[\frac{4a(1-a)\phi}{4a(1-a)(\sigma\phi-1)+1}\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}\right]}{\left[\sigma + \eta - \frac{2(1-a)(\sigma-1)}{2a(\phi-1)+1}\right]}, \quad \text{and}
$$

$$
\delta_3 \equiv \frac{2a(\sigma\phi-1)+1}{2a\phi\omega}.
$$

The difference rule Equation (A24) can be rewritten as:

$$
\kappa(\gamma_{Lt} - \gamma^*_{Ut}) = \delta_1(\tilde{Y}_{Lt} - \tilde{Y}_{Ut}) + \delta_2\tilde{\mathcal{W}}_t. \tag{A25}
$$

Combining Equation (A22) with the UIP condition (15), the FOC for f^*_t (A23), and the difference rule (A25),

$$
\begin{aligned}
0 = & -2a\phi\omega(f^*_t - u_t) \\
& + [2a(\sigma\phi-1)+1]\kappa[(E_t\gamma_{Lt+1} - \gamma_{Lt}) - (E_t\gamma^*_{Ut+1} - \gamma^*_{Ut})], \\
f^*_t = & \, \delta_3\kappa[(E_t\gamma_{Lt+1} - \gamma_{Lt}) - (E_t\gamma^*_{Ut+1} - \gamma^*_{Ut})] + u_t \\
= & \, \delta_3[\delta_1\{(E_t\tilde{Y}_{Lt+1} - \tilde{Y}_{Lt}) - (E_t\tilde{Y}_{Ut+1} - \tilde{Y}_{Ut})\} + \delta_2(\tilde{\mathcal{W}}_{t+1} - \tilde{\mathcal{W}}_t)] + u_t, \\
f^*_t = & \, \frac{\delta_1\delta_3}{1 - \delta_2\delta_3\omega}\{(E_t\tilde{Y}_{Lt+1} - \tilde{Y}_{Lt}) - (E_t\tilde{Y}_{Ut+1} - \tilde{Y}_{Ut})\} + u_t.
\end{aligned}
$$

This gives the optimal FXI rule as in Equation (16), where $\xi_Y \equiv \delta_1\delta_3/(1 - \delta_2\delta_3\omega)$.

Part (ii). First, consider the case without FXI. Equation (7) implies that $\partial\tilde{\mathcal{W}}_t/\partial\hat{A}_t > 0$.

Moreover, from Equation (A10),

$$\frac{\partial \tilde{\mathcal{T}}_t}{\partial \tilde{\mathcal{W}}_t} = -\frac{2a-1}{4a(1-a)(\sigma\phi-1)+1} < 0. \tag{A26}$$

The NKPC for local firms, combined with $\pi_{Lt} = 0$ for all t (Equation (11)), gives $\partial \tilde{\mathcal{W}}_t / \partial \tilde{Y}_{Lt} > 0$ and $\partial \tilde{Y}_{Lt} / \partial \hat{A}_t < 0$. Similarly, $\partial \tilde{Y}_{Ut} / \partial \hat{A}_t > 0$.

Next, consider the case with FXI. The optimal FXI rule Equation (16) gives $\partial \tilde{f}^*_t / \partial \hat{A}_t > 0$ and the UIP condition Equation (15) gives $\partial \tilde{W}_t / \partial \hat{f}^*_t < 0$. Hence,

$$\left(\frac{\partial \tilde{\mathcal{W}}_t}{\partial A_t}\right)^{\text{No FXI}} > \left(\frac{\partial \tilde{\mathcal{W}}_t}{\partial A_t}\right)^{\text{FXI}}.$$

This implies Equation (19). Moreover, since $\partial \tilde{\mathcal{W}}_t / \partial \tilde{Y}_{Lt} > 0$, we obtain:

$$\left(\frac{\partial \tilde{Y}_{Lt}}{\partial \hat{A}_t}\right)^{\text{No FXI}} < \left(\frac{\partial \tilde{Y}_{Lt}}{\partial \hat{A}_t}\right)^{\text{FXI}}$$

for the local economy, and similarly,

$$\left(\frac{\partial \tilde{Y}_{Ut}}{\partial \hat{A}_t}\right)^{\text{No FXI}} > \left(\frac{\partial \tilde{Y}_{Ut}}{\partial \hat{A}_t}\right)^{\text{FXI}}$$

for the US. This gives Equation (17) and Equation (18). This completes the proof. □

A.4 Proof of Proposition 2

In this section, I first discuss the optimal monetary policy rule when FXI is not available, following Corsetti et al. (2023). Next, I consider the case where both monetary policy and FXI are set optimally and derive the optimal policy rules in Appendix A.4.

A.4.1 Optimal Monetary Policy when FXI is Not Available (Corsetti et al. 2023)

Under cooperation and commitment, the central banks in the two countries minimize the loss (13) subject to the NKPCs (8) and (9) and the UIP condition (15). The first-order conditions (FOCs) are given by Equations (A20)-(A23) and:

$$\pi_{Lt}: \quad 0 = -\frac{\theta}{\kappa}\pi_{Lt} - \gamma_{Lt} + \gamma_{Lt-1}, \tag{A27}$$

$$\pi^*_{Ut}: \quad 0 = -\frac{\theta}{\kappa}\pi^*_{Ut} - \gamma^*_{Ut} + \gamma^*_{Ut-1}. \tag{A28}$$

Under the assumption of $f^*_t = u_t = 0$, we have $E_t\tilde{\mathcal{W}}_{t+1} - \tilde{\mathcal{W}}_t = 0$. By taking the sum of the FOCs for the output gap in the two countries and combining it with the FOCs for inflation, the optimal rule can be expressed in terms of the sum of world inflation and output gaps (the sum rule):

$$\begin{aligned} 0 &= \tilde{Y}_{Lt} + \tilde{Y}_{Ut} + \theta(p_{Lt} + p^*_{Ut}) \\ &= (\tilde{Y}_{Lt} - \tilde{Y}_{Lt-1}) + (\tilde{Y}_{Ut} - \tilde{Y}_{Ut-1}) + \theta(\pi_{Lt} + \pi^*_{Ut}). \end{aligned} \tag{A29}$$

Next, taking the difference between the FOCs (A20) and (A21) and combining it with the FOCs for inflation (the difference rule),

$$\begin{aligned} 0 = &\left[\sigma + \eta - \frac{4a(1-a)(\sigma\phi-1)\sigma}{4a(1-a)(\sigma\phi-1)+1}\right](\tilde{Y}_{Lt} - \tilde{Y}_{Ut}) \\ &+ \frac{4a(1-a)\phi}{4a(1-a)(\sigma\phi-1)+1}\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}\tilde{\mathcal{W}}_t \\ &+ \left[\sigma + \eta - \frac{2(1-a)(\sigma-1)}{2a(\phi-1)+1}\right]\theta(\tilde{p}_{Lt} - \tilde{p}^*_{Ut}) \\ &+ 2\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}(\lambda_t - \beta^{-1}\lambda_{t-1}). \end{aligned} \tag{A30}$$

The FOC for the net foreign asset implies:

$$-(\lambda_t - \beta^{-1}\lambda_{t-1}) = (1-a)\frac{2a(\sigma\phi-1)+1}{4a(1-a)(\sigma\phi-1)+1}\theta(p_{Lt} - p^*_{Ut}).$$

Combining the sum and difference rules with the FOC for the net foreign asset,

$$\begin{aligned} 0 = &\left[\sigma + \eta - \frac{4a(1-a)(\sigma\phi-1)\sigma}{4a(1-a)(\sigma\phi-1)+1}\right][(\tilde{Y}_{Lt} - \tilde{Y}_{Lt-1}) - (\tilde{Y}_{Ut} - \tilde{Y}_{Ut-1}) + \theta(\pi_{Lt} - \pi^*_{Ut})] \\ &+ \frac{4a(1-a)\phi}{4a(1-a)(\sigma\phi-1)+1}\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}(\tilde{\mathcal{W}}_t - \tilde{\mathcal{W}}_{t-1}). \end{aligned}$$

Rearranging the terms, the optimal rule can be expressed in terms of the difference of inflation and output gaps across countries:

$$0 = (\tilde{Y}_{Lt} - \tilde{Y}_{Lt-1}) - (\tilde{Y}_{Ut} - \tilde{Y}_{Ut-1}) + \theta(\pi_{Lt} - \pi^*_{Ut})$$

$$+\frac{4a(1-a)\phi}{\sigma+\eta[4a(1-a)(\sigma\phi-1)+1]}\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}(\tilde{\mathcal{W}}_t-\tilde{\mathcal{W}}_{t-1}).$$

Combining the sum and difference rules, we obtain the country-specific monetary policy rules (20) and (21), where:

$$\xi_{\mathcal{W}}\equiv\frac{2a(1-a)\phi}{\sigma+\eta[4a(1-a)(\sigma\phi-1)+1]}\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}.$$

A.4.2 Optimal Monetary Policy and FXI

When both monetary policy and FXI are set optimally, the optimality conditions are given by Equations (A20) to (A23). First, the sum rule of inflation and the output gap can be characterized by Equation (A29) similarly to the case without FXI. Next, to derive the difference rule, combining the FOCs for the output gap (Equation (A30)) and FXI (Equation (A23)),

$$\begin{aligned}
0 &= \left[\sigma+\eta-\frac{4a(1-a)(\sigma\phi-1)\sigma}{4a(1-a)(\sigma\phi-1)+1}\right](\tilde{Y}_{Lt}-\tilde{Y}_{Ut})\\
&\quad+\frac{4a(1-a)\phi}{4a(1-a)(\sigma\phi-1)+1}\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}\tilde{\mathcal{W}}_t\\
&\quad+\left[\sigma+\eta-\frac{2(1-a)(\sigma-1)}{2a(\phi-1)+1}\right]\theta(\tilde{p}_{Lt}-\tilde{p}^*_{Ut})\\
&=\left[\sigma+\eta-\frac{4a(1-a)(\sigma\phi-1)\sigma}{4a(1-a)(\sigma\phi-1)+1}\right][\tilde{Y}_{Lt}-\tilde{Y}_{Ut}+\theta(\tilde{p}_{Lt}-\tilde{p}^*_{Ut})]\\
&\quad+2(1-a)\frac{2a(\sigma\phi-1)+1}{4a(1-a)(\sigma\phi-1)+1}\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}\theta(p_{Lt}-p^*_{Ut})\\
&\quad+\frac{4a(1-a)\phi}{4a(1-a)(\sigma\phi-1)+1}\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}\tilde{\mathcal{W}}_t
\end{aligned}$$

Taking the first difference and rearranging the terms, the difference rule can be written as:

$$\begin{aligned}
0 &= (\tilde{Y}_{Lt}-\tilde{Y}_{Lt-1})-(\tilde{Y}_{Ut}-\tilde{Y}_{Ut-1})+\theta(\pi_{Lt}-\pi^*_{Ut})\\
&\quad+2(1-a)\frac{2a(\sigma\phi-1)+1}{\sigma+\eta\{4a(1-a)(\sigma\phi-1)+1\}}\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}\theta(\pi_{Lt}-\pi^*_{Ut})\\
&\quad+\frac{4a(1-a)\phi}{\sigma+\eta\{4a(1-a)(\sigma\phi-1)+1\}}\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}(\tilde{\mathcal{W}}_t-\tilde{\mathcal{W}}_{t-1}).
\end{aligned}\tag{A31}$$

To further rewrite Equation (A31), combining Equations (A22) and (A23) gives:

$$0=2a\phi\tilde{\mathcal{W}}_t-[2a(\sigma\phi-1)+1]\theta(p_{Lt}-p^*_{Ut}).$$

Taking the first difference and rearranging the terms,

$$\theta(\pi_{Lt} - \pi_{Ut}^*) = \frac{2a\phi}{2a(\sigma\phi - 1) + 1}(\tilde{\mathcal{W}}_t - \tilde{\mathcal{W}}_{t-1}). \tag{A32}$$

Combining Equations (A31) and (A32), the difference rule can be rewritten as:

$$\begin{aligned} 0 =& (\tilde{Y}_{Lt} - \tilde{Y}_{Lt-1}) - (\tilde{Y}_{Ut} - \tilde{Y}_{Ut-1}) + \theta(\pi_{Lt} - \pi_{Ut}^*) \\ &+ \frac{8a(1-a)\phi}{\sigma + \eta\{4a(1-a)(\sigma\phi - 1) + 1\}} \frac{2a(\sigma\phi - 1) + 1 - \sigma}{2a(\phi - 1) + 1}(\tilde{\mathcal{W}}_t - \tilde{\mathcal{W}}_{t-1}). \end{aligned} \tag{A33}$$

Using Equations (A29) and (A33), we obtain the optimal monetary policy rules as Equations (20) and (21), where:

$$\xi_W = \frac{4a(1-a)\phi}{\sigma + \eta\{4a(1-a)(\sigma\phi - 1) + 1\}} \frac{2a(\sigma\phi - 1) + 1 - \sigma}{2a(\phi - 1) + 1}.$$

The optimal FXI rule can be characterized similarly to Appendix A.4.1 using Equations (A20)-(A22). □

A.5 Details on Calibration in Section 4

This section provides details on the calibration of the quantitative model. The key parameter is the degree of currency market segmentation ω, which is determined by the risk aversion of financial intermediaries. I calibrate this parameter to match data on FXI and UIP deviation for 11 major currencies.

I use data on 11 major economies: Australia, Brazil, Canada, China, the Euro area, India, Japan, Korea, Russia, Switzerland, and the United Kingdom. The data is quarterly and the sample period is 2000-23. The UIP deviation is constructed using data on the spot exchange rate, quarterly exchange rate forecast, and interbank rate available in Bloomberg. The FXI data is collected from multiple sources, including central bank websites, FRED, and IMF data (Adler et al. 2023).

Table A1 shows the summary statistics of FXI volume in terms of billions of US dollars (panel a) and the percentage ratio over GDP (panel b) in the sample countries. The median volume of interventions is 0.18% of GDP for dollar sales and 0.35% of GDP for dollar purchases, respectively.

To identify the effect of FXI, I exploit deviations from an estimated FXI policy rule. The

Table A1: Summary Statistics for FXI

(a) FXI Volume (Billions of US Dollars)

	Mean	Median	SD	p25	p75	p90	Max	Obs
Sell Dollars (Billions)	10.19	2.09	24.65	0.74	7.73	22.10	179.88	262
Buy Dollars (Billions)	14.66	5.53	25.71	1.59	13.21	33.79	164.17	448

(b) FXI over GDP Ratio (Percentage)

	Mean	Median	SD	p25	p75	p90	Max	Obs
Sell Dollars (% GDP)	0.48	0.18	0.92	0.06	0.52	1.09	9.11	262
Buy Dollars (% GDP)	0.91	0.35	1.79	0.13	1.02	2.08	19.86	448

Note: The table shows the summary statistics for FXI in the sample. Panel (a) shows the volume of FXI in terms of billions of US dollars. Panel (b) shows the FXI over GDP ratio in percentage terms. In each panel, the first row shows interventions by selling the US dollar and buying the local currency and the second row shows interventions by buying the US dollar and selling the local currency.

idea is to extract components of interventions that cannot be extracted by factors such as past intervention and exchange rate movements. The idea is a conventional approach in the literature to minimizing the endogeneity of FXI (Fatum and Hutchison 2010, Kuersteiner et al. 2018, Fratzscher et al. 2019, Rodnyansky et al. 2024) and analogous to estimating residuals from a monetary policy Taylor rule (Taylor 1993, Maddaloni and Peydró 2011, Dell'Ariccia et al. 2017).

In particular, I estimate the following equation:

$$FXI_{i,t} = \alpha + \beta X_{i,t-1} + \gamma_i + \epsilon_{i,t}. \tag{A34}$$

$FXI_{c,t}$ is the FXI over GDP ratio in country i and period t. A positive (negative) value implies the net purchases (sales) of dollars against the local currency. $X_{i,t-1}$ is the set of lagged control variables, including the past FXI over GDP ratio, trend and volatility of the spot exchange rate, UIP deviation, VIX index, local and US monetary policy rate, CPI inflation, unemployment rate, and current account over GDP ratio. γ_i is the country fixed effect.

Table A2 shows the estimation result. Column (1) shows the result including all sample countries. The estimate shows that around 18% of the variation in FXI can be explained by the set of controls ($R^2 = 0.176$) while the remaining 82% are unexplained. As a robustness check, column (2) excludes small-open economies (Australia, Canada, Korea, and Switzerland) and a country with managed exchange rate regime (China). Around 26% of the variation in FXI can be explained while the remaining 74% are unexplained. Consistent with Fratzscher et al. (2019) and Rodnyansky et al. (2024), observed controls such as past exchange rates have limited explanatory power for FXI. I define the intervention to be unexpected (expected) if the residual from estimating Equation (A34) is larger (smaller) than the median in absolute value.

Having shown that the variation in FXI is difficult to predict, I study the effect of FXI on the UIP deviation. I estimate the following equation:

$$UIP_{i,t} - UIP_{i,t-1} = \alpha + \beta FXI_{i,t} + \gamma_i + \epsilon_{i,t}, . \tag{A35}$$

where $UIP_{i,t}$ is the expected currency excess return against the dollar. $UIP_{i,t}$ is in terms of percentage points, and a positive value implies that the local currency has a higher return on investment than the US dollar.

Table A3 show the result. Columns (1) and (2) show the results using all sample countries. Column (1) uses all intervention episodes in the sample. Column (2) excludes "expected" interventions as defined in the first-step regression for identification of FXI. The result shows that a net purchase of dollars corresponding to 1% of GDP increases the UIP deviation by around 0.51-0.59 percentage points. The result of column (2) is used for the impulse response analyses.

As a robustness check, columns (3) and (4) conduct a similar exercise after excluding small-open economies and economies with managed exchange rate regimes. A net purchase of dollars corresponding to 1% of GDP increases the UIP deviation by around 1.1-1.6 percentage points. A possible reason why FXI is more effective is that the sample of small open economies includes those with liquid financial markets, such as Switzerland.

As discussed in Maggiori (2022), an important challenge in the literature is to estimate the degree of currency market segmentation and identify the causal effect of capital flows on exchange rates. Koijen and Yogo (2024) estimate a model of asset demand system using bilateral international portfolio holdings data and decompose the variation in exchange rates into portfolio flows and shifts in asset demand. Camanho et al. (2022) and Bippus et al. (2024) use granular portfolio holdings data for mutual funds and global banks, respectively, and use an instrumental variable approach to estimate causal effects of capital flows on exchange rates.

Another important challenge is the causal identification of the effect of FXI on exchange rates and the economy. Rodnyansky et al. (2024) combine daily intervention data with high-

Table A2: First-Step: FXI Policy Rule

Dependent Variable	Net \$ Purchases / GDP (%)	
	(1)	(2)
Lagged FXI / GDP (%)	0.129	0.283^{***}
	(0.135)	(0.092)
Lagged Exchange Depreciation (%)	-0.005	-0.006
	(0.014)	(0.009)
Lagged Exchange Volatility (%)	-0.089	-0.006
	(0.067)	(0.033)
Lagged UIP Deviation (p.p.)	-0.010^{*}	-0.012^{*}
	(0.006)	(0.006)
Lagged log(VIX)	0.135	0.065
	(0.154)	(0.118)
Lagged Policy Rate (Local)	0.078^{**}	0.030
	(0.031)	(0.025)
Lagged Policy Rate (US)	-0.041	0.077^{*}
	(0.054)	(0.045)
Lagged CPI Inflation (%)	-0.041	-0.026
	(0.042)	(0.053)
Lagged Unemployment Rate (%)	-0.004	0.003
	(0.029)	(0.021)
Lagged Current Account / GDP (%)	0.115^{**}	0.069^{*}
	(0.052)	(0.038)
R^2	0.176	0.259
N	627	309
Country Fixed Effect	✓	✓
Exclude Small Economy		✓
Exclude Managed Exchange Rate		✓

Note: The table shows the result of the first-step regression in Equation (A34). The dependent variable is the percentage ratio of FXI over GDP and a positive value indicates a purchase of dollars and a sale of the local currency. Column (1) includes all sample economies. Column (2) excludes small-open economies (Australia, Canada, Korea, and Switzerland) and a country with managed exchange rate regime (China). Standard errors are in parentheses. Standard errors are clustered at the quarter level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

Table A3: Second-Step: The Effect of FXI on UIP Deviation

Dependent Variable	$UIP_t - UIP_{t-1}$			
	(1)	(2)	(3)	(4)
Net \$ Purchases / GDP (%)	0.589**	0.509**	1.599***	1.106***
	(0.264)	(0.212)	(0.182)	(0.178)
R^2	0.004	0.013	0.009	0.020
N	706	392	367	212
Country Fixed Effect	✓	✓	✓	✓
Identified Intervention		✓		✓
Exclude Small Economy			✓	✓
Exclude Managed Exchange Rate			✓	✓

Note: The table shows the result of the second-step regression in Equation (A34). The dependent variable is the change in the UIP deviation from period $t-1$ to t and the positive value implies that the local currency has a higher return than the US dollar. Columns (1) and (2) include all sample economies. Column (1) includes all intervention episodes. Column (2) excludes expected interventions, defined as cases where the residual from estimating Equation (A34) is smaller than the median. Columns (3) and (4) conduct similar exercises after excluding small-open economies (Australia, Canada, Korea, and Switzerland) and a country with managed exchange rate regime (China). Standard errors are clustered at the quarter level. The symbols *, **, and *** indicate significance at 10%, 5%, and 1% levels, respectively.

frequency US monetary shock, daily exchange rates, and firm-level stock price and balance sheet data and study heterogeneous effects of FXI across firms.

Quantitatively, the optimal volume of FXI is smaller under markup shocks (a dollar purchase equivalent to 0.4-1.1% of GDP) than productivity shocks (a dollar sales equivalent to 0.01-0.03% of GDP). This is because productivity shocks affect inflation only indirectly via changes in the risk-sharing wedge, while markup shocks affect inflation directly by increasing the marginal cost.

A.6 Proof of Proposition 3

Under cooperation and commitment, the central banks in the two countries minimize the loss (25) subject to the NKPCs (23) and (24) for the local economy and (9) for the US, the UIP condition (15), and the relationship linking the price dispersion to the relative inflation rates and terms of trade across countries (Corsetti et al. 2023):

$$\pi_{Ut} - \pi_{Lt} = \tilde{\Delta}_t - \tilde{\Delta}_{t-1} + \tilde{\mathcal{T}}_t - \tilde{\mathcal{T}}_{t-1}. \tag{A36}$$

Let γ_t be the Lagrangian multiplier on Equation (A36). Assuming $\pi_{Lt} = \pi^*_{Ut} = 0$ for all t, the first-order conditions can be written as:

$$
\begin{aligned}
\tilde{Y}_{Lt}: \quad 0 = & -(\sigma+\eta)\tilde{Y}_{Lt} + \frac{2a(1-a)(\sigma\phi-1)\sigma}{4a(1-a)(\sigma\phi-1)+1}(\tilde{Y}_{Lt}-\tilde{Y}_{Ut}) \\
& - \frac{2a(1-a)\phi}{4a(1-a)(\sigma\phi-1)+1}\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}(\tilde{\mathcal{W}}_t + \tilde{\Delta}_{Lt}) \\
& + \left[\sigma+\eta-\frac{(1-a)(\sigma-1)}{2a(\phi-1)+1}\right]\kappa(\gamma_{Lt}+\gamma^*_{Lt}) + \frac{(1-a)(\sigma-1)}{2a(\phi-1)+1}\kappa\gamma^*_{Ut} \\
& + \frac{1}{2a(\phi-1)+1}(\beta E_t\gamma_{t+1}-\gamma_t) \\
& - \frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}(\lambda_t - \beta^{-1}\lambda_{t-1}),
\end{aligned} \tag{A37}
$$

$$
\begin{aligned}
\tilde{Y}_{Ut}: \quad 0 = & -(\sigma+\eta)\tilde{Y}_{Ut} - \frac{2a(1-a)(\sigma\phi-1)\sigma}{4a(1-a)(\sigma\phi-1)+1}(\tilde{Y}_{Lt}-\tilde{Y}_{Ut}) \\
& + \frac{2a(1-a)\phi}{4a(1-a)(\sigma\phi-1)+1}\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}(\tilde{\mathcal{W}}_t + \tilde{\Delta}_{Lt}) \\
& + \left[\sigma+\eta-\frac{(1-a)(\sigma-1)}{2a(\phi-1)+1}\right]\kappa\gamma^*_{Ut} + \frac{(1-a)(\sigma-1)}{2a(\phi-1)+1}\kappa(\gamma_{Lt}+\gamma^*_{Lt}) \\
& - \frac{1}{2a(\phi-1)+1}(\beta E_t\gamma_{t+1}-\gamma_t) \\
& + \frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}(\lambda_t - \beta^{-1}\lambda_{t-1}),
\end{aligned} \tag{A38}
$$

$$
\begin{aligned}
\tilde{\mathcal{B}}_t: \quad 0 = & -\frac{2a(1-a)\phi}{4a(1-a)(\sigma\phi-1)+1}(E_t\tilde{\mathcal{W}}_{t+1} - \tilde{\mathcal{W}}_t) \\
& + (1-a)\frac{2a(\sigma\phi-1)+1}{4a(1-a)(\sigma\phi-1)+1} \\
& \times \kappa[(E_t(\gamma_{Lt+1}+\gamma^*_{Lt+1}) - (\gamma_{Lt}+\gamma^*_{Lt})) - (E_t\gamma^*_{Ut+1} - \gamma^*_{Ut})] \\
& - \frac{2a-1}{4a(1-a)(\sigma\phi-1)+1}[(\beta E_t\gamma_{t+2} - E_t\gamma_{t+1}) - (\beta E_t\gamma_{t+1} - \gamma_t)] \\
& - [(E_t\lambda_{t+1} - \beta^{-1}\lambda_t) - (\lambda_t - \beta^{-1}\lambda_{t-1})],
\end{aligned} \tag{A39}
$$

$$
f_t: \quad \lambda_t = 0. \tag{A40}
$$

For analytical tractability, I assume that the FOC in $\tilde{\Delta}_t$ only takes into account the terms-of-trade relationship (A36):

$$
\gamma_t = -\delta_4\tilde{\Delta}_t, \quad \text{where} \quad \delta_4 \equiv \frac{4a(1-a)\omega}{2a-1}.
$$

Define $\gamma_{\Delta t} \equiv 2(\beta E_t \gamma_{t+1} - \gamma_t)$. Taking the difference between Equations (A37) and (A38),

$$\begin{aligned} \frac{1}{2a(\phi-1)+1}\gamma_{\Delta t} = & \left[\sigma + \eta - \frac{4a(1-a)(\sigma\phi-1)\sigma}{4a(1-a)(\sigma\phi-1)+1}\right](\tilde{Y}_{Lt} - \tilde{Y}_{Ut}) \\ & + \frac{4a(1-a)\phi}{4a(1-a)(\sigma\phi-1)+1}\frac{2a(\sigma\phi-1)+1-\sigma}{2a(\phi-1)+1}(\tilde{\mathcal{W}}_t + \tilde{\Delta}_{Lt}) \\ & - \left[\sigma + \eta - \frac{2(1-a)(\sigma-1)}{2a(\phi-1)+1}\right]\kappa(\gamma_{Lt} + \gamma_{Lt}^* - \gamma_{Ut}^*). \end{aligned} \tag{A41}$$

I define δ_5 and δ_6 as:

$$\delta_5 \equiv \frac{1}{[2a(\phi-1)+1]\left[\sigma+\eta-\frac{2(1-a)(\sigma-1)}{2a(\phi-1)+1}\right]} \quad \text{and} \quad \delta_6 \equiv \frac{2a-1}{4a(1-a)\phi\omega}.$$

The difference rule Equation (A41) can be rewritten as:

$$\kappa(\gamma_{Lt} + \gamma_{Lt}^* - \gamma_{Ut}^*) = \delta_1(\tilde{Y}_{Lt} - \tilde{Y}_{Ut}) + \delta_2\tilde{\mathcal{W}}_t - \delta_5\gamma_{\Delta t}. \tag{A42}$$

Combining Equation (A39) with the UIP condition (15) and the difference rule (A42),

$$\begin{aligned} 0 = & -2a(1-a)\phi\omega(f_t^* - u_t) \\ & + (1-a)[2a(\sigma\phi-1)+1]\kappa[(E_t\gamma_{Lt+1} + E_t\gamma_{Lt+1}^*) - (\gamma_{Lt} + \gamma_{Lt}^*) - (E_t\gamma_{Ut+1}^* - \gamma_{Ut}^*)] \\ & - (2a-1)[(\beta E_t\gamma_{t+2} - E_t\gamma_{t+1}) - (\beta E_t\gamma_{t+1} - \gamma_t)], \\ f_t^* = & \ \delta_3\kappa[(E_t\gamma_{Lt+1} + E_t\gamma_{Lt+1}^*) - (\gamma_{Lt} + \gamma_{Lt}^*) - (E_t\gamma_{Ut+1}^* - \gamma_{Ut}^*)] \\ & - \delta_6[(\beta E_t\gamma_{t+2} - E_t\gamma_{t+1}) - (\beta E_t\gamma_{t+1} - \gamma_t)] + u_t \\ = & \ \delta_3\left[\begin{array}{l} \delta_1\{(E_t\tilde{Y}_{Lt+1} - \tilde{Y}_{Lt}) - (E_t\tilde{Y}_{Ut+1} - \tilde{Y}_{Ut})\} \\ +\delta_2(E_t\tilde{\mathcal{W}}_{t+1} - \tilde{\mathcal{W}}_t + E_t\tilde{\Delta}_{Lt+1} - \tilde{\Delta}_{Lt}) - \delta_5(E_t\gamma_{\Delta Lt+1} - \gamma_{\Delta Lt}) \end{array}\right] \\ & - \delta_6(E_t\gamma_{\Delta Lt+1} - \gamma_{\Delta Lt}) + u_t \\ = & \ \delta_1\delta_3\{(E_t\tilde{Y}_{Lt+1} - \tilde{Y}_{Lt}) - (E_t\tilde{Y}_{Ut+1} - \tilde{Y}_{Ut})\} + \delta_2\delta_3(E_t\tilde{\mathcal{W}}_{t+1} - \tilde{\mathcal{W}}_t + E_t\tilde{\Delta}_{Lt+1} - \tilde{\Delta}_{Lt}) \\ & - (\delta_3\delta_5 + \delta_6)(E_t\gamma_{\Delta Lt+1} - \gamma_{\Delta Lt}) + u_t \\ = & \ \delta_1\delta_3\{(E_t\tilde{Y}_{Lt+1} - \tilde{Y}_{Lt}) - (E_t\tilde{Y}_{Ut+1} - \tilde{Y}_{Ut})\} + \delta_2\delta_3(\omega(f_t^* - u_t) + E_t\tilde{\Delta}_{Lt+1} - \tilde{\Delta}_{Lt}) \\ & - (\delta_3\delta_5 + \delta_6)\delta_4(E_t\tilde{\Delta}_{Lt+1} - \tilde{\Delta}_{Lt}) + u_t. \end{aligned}$$

Rearranging the terms,

$$
\begin{aligned}
f_t^* =& \frac{\delta_1\delta_3}{1-\delta_2\delta_3\omega}\{(E_t\tilde{Y}_{Lt+1}-\tilde{Y}_{Lt})-(E_t\tilde{Y}_{Ut+1}-\tilde{Y}_{Ut})\} \\
&+ \frac{\delta_2\delta_3+(\delta_3\delta_5+\delta_6)\delta_4}{1-\delta_2\delta_3\omega}(E_t\tilde{\Delta}_{Lt+1}-\tilde{\Delta}_{Lt})+u_t.
\end{aligned}
$$

This gives the optimal FXI rule under DCP as in Equation (26), where $\xi_Y^{DCP} \equiv \delta_1\delta_3/(1-\delta_2\delta_3\omega)$ and $\xi_{\mathcal{W}}^{DCP} \equiv [\delta_2\delta_3+(\delta_3\delta_5+\delta_6)\delta_4]/(1-\delta_2\delta_3\omega)$.

A.7 Proof of Proposition 4

Under inflation-targeting monetary policy ($\pi_{Lt} = 0$ for all t), the local central bank's objective function Equation (27) reduces to:

$$
\mathcal{L} = -E_0\sum_{t=0}^{\infty}\beta^t\tilde{Y}_{Lt}^2. \tag{A43}
$$

The local central bank minimizes the loss (A43) subject to the NKPC (8) and the UIP condition (15). Let γ_{Lt} and λ_t be the Lagrangian multiplier on the NKPC and UIP condition, respectively. The FOCs can be written as:

$$
\tilde{Y}_{Lt}: \quad 0 = -(\sigma+\eta)\tilde{Y}_{Lt} + \left[\sigma+\eta-\frac{(1-a)(\sigma-1)}{2a(\phi-1)+1}\right]\kappa\gamma_{Lt}, \tag{A44}
$$

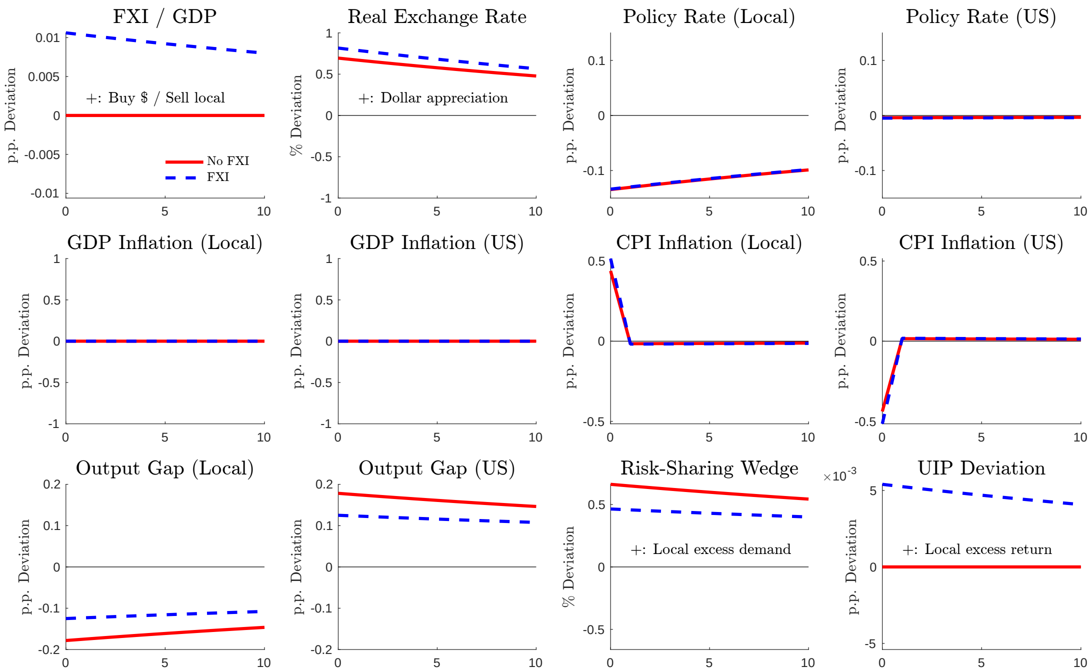
$$
\begin{aligned}
\tilde{\mathcal{B}}_t: \quad 0 =& (1-a)\frac{2a(\sigma\phi-1)+1}{4a(1-a)(\sigma\phi-1)+1}\times\kappa(E_t\gamma_{Lt+1}-\gamma_{Lt}) \\
&-[(E_t\lambda_{t+1}-\beta^{-1}\lambda_t)-(\lambda_t-\beta^{-1}\lambda_{t-1})],
\end{aligned} \tag{A45}
$$

$$
f_t: \quad \lambda_t = 0. \tag{A46}
$$

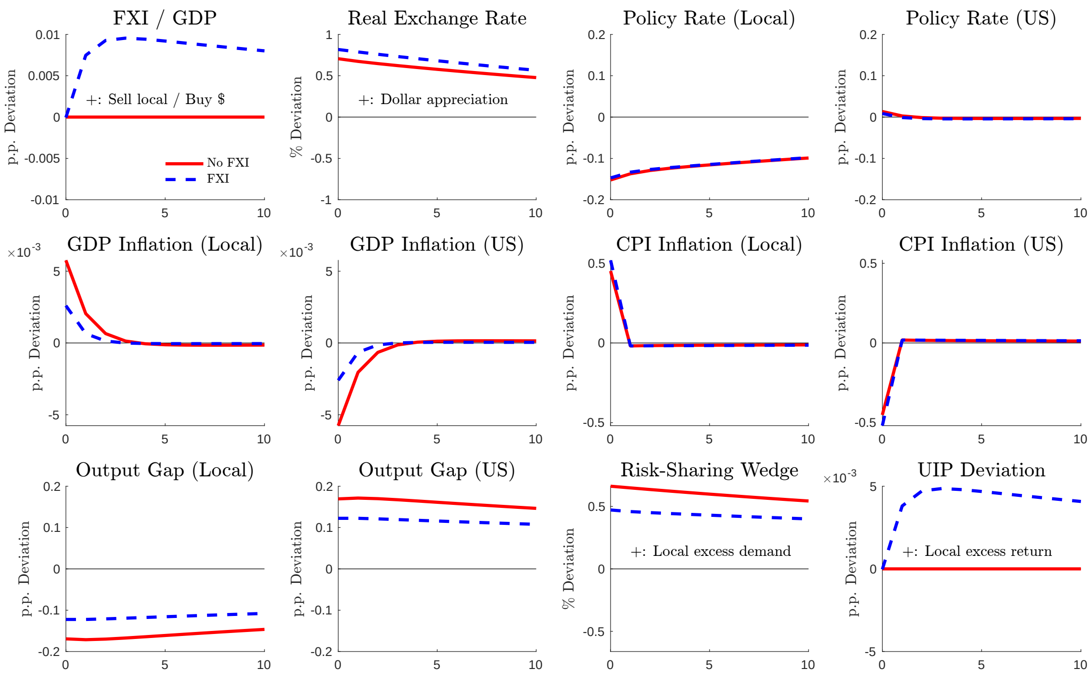
Combining Equations (A44)-(A46), we obtain the optimal FXI rule (28). □

Figure A1: Impulse Responses to a Local Productivity Increase

(a) Optimal FXI under Inflation-Targeting Monetary Policy



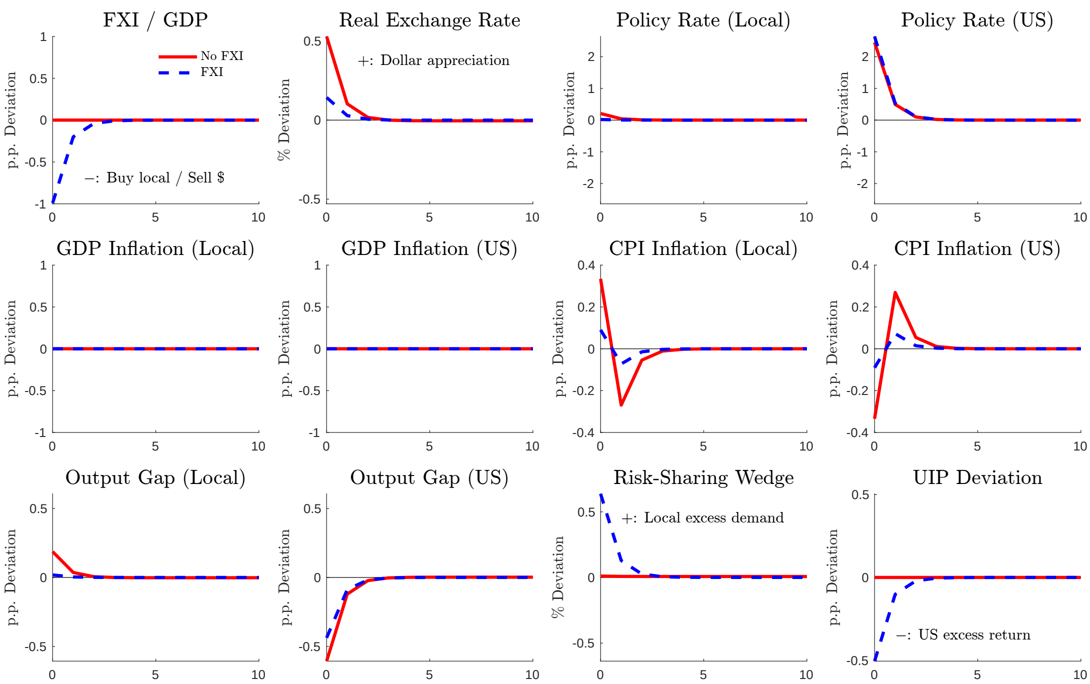
(b) Optimal Monetary Policy and FXI



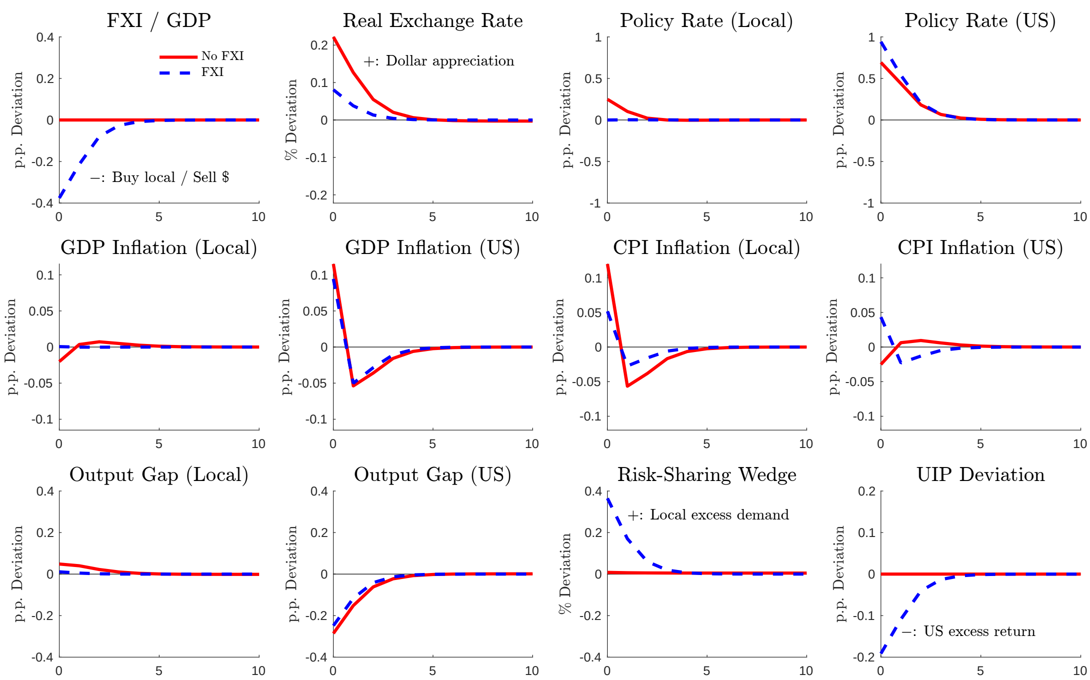
Note: This figure plots impulse responses to a one-standard-deviation increase in local productivity. Panel (a) shows the case where monetary policy targets inflation and FXI is set optimally. Panel (b) shows the case where both monetary policy and FXI are set optimally. The red and blue lines show the results under no intervention and cooperative intervention cases, respectively.

Figure A2: Impulse Responses to a US Cost-Push Inflation

(a) Optimal FXI under Inflation-Targeting Monetary Policy



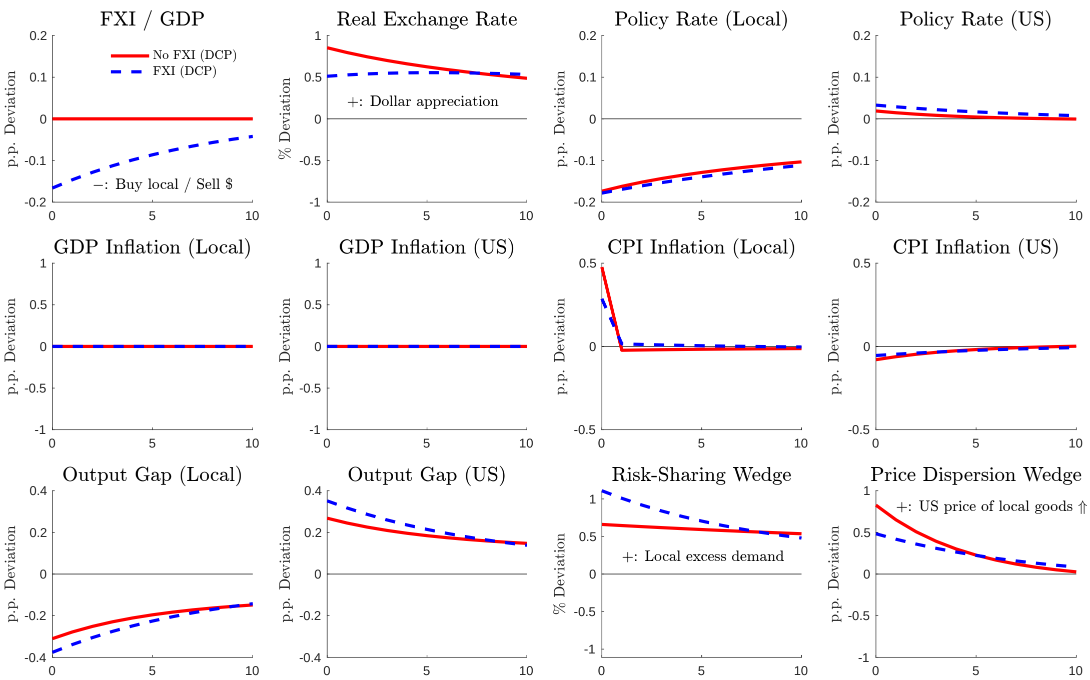
(b) Optimal Monetary Policy and FXI



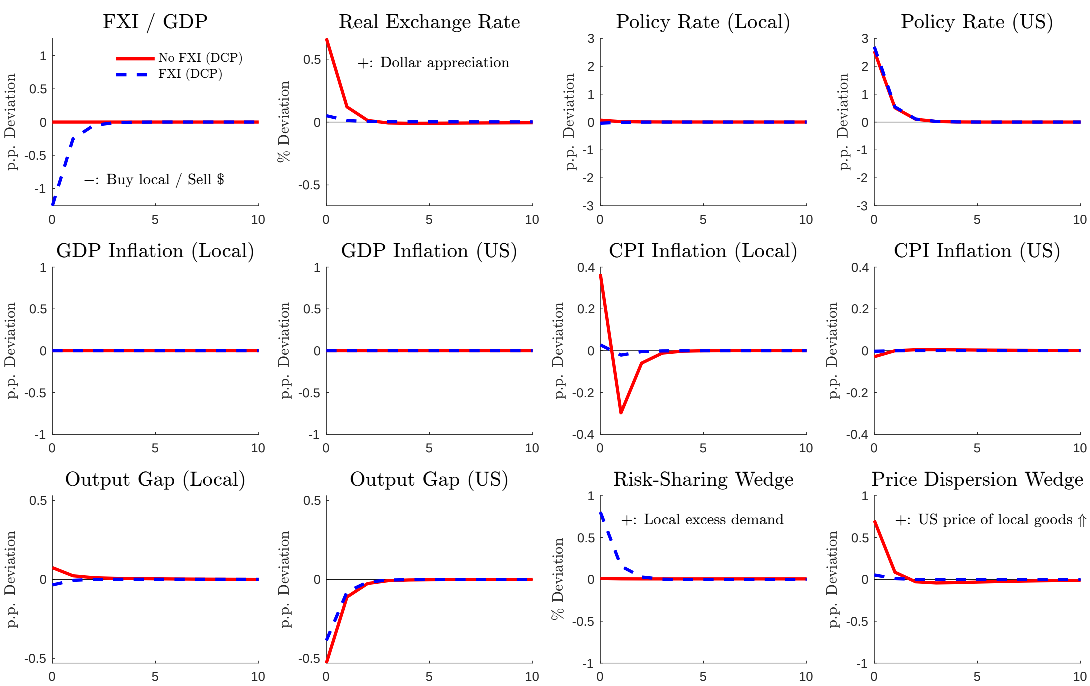
Note: This figure plots impulse responses to a one-standard-deviation increase in US markup. Panel (a) shows the case where monetary policy targets inflation and FXI is set optimally. Panel (b) shows the case where both monetary policy and FXI are set optimally. The red and blue lines show the results under no intervention and cooperative intervention cases, respectively.

Figure A3: Impulse Responses (Dominant Currency Pricing)

(a) Impulse Responses to a Local Productivity Increase

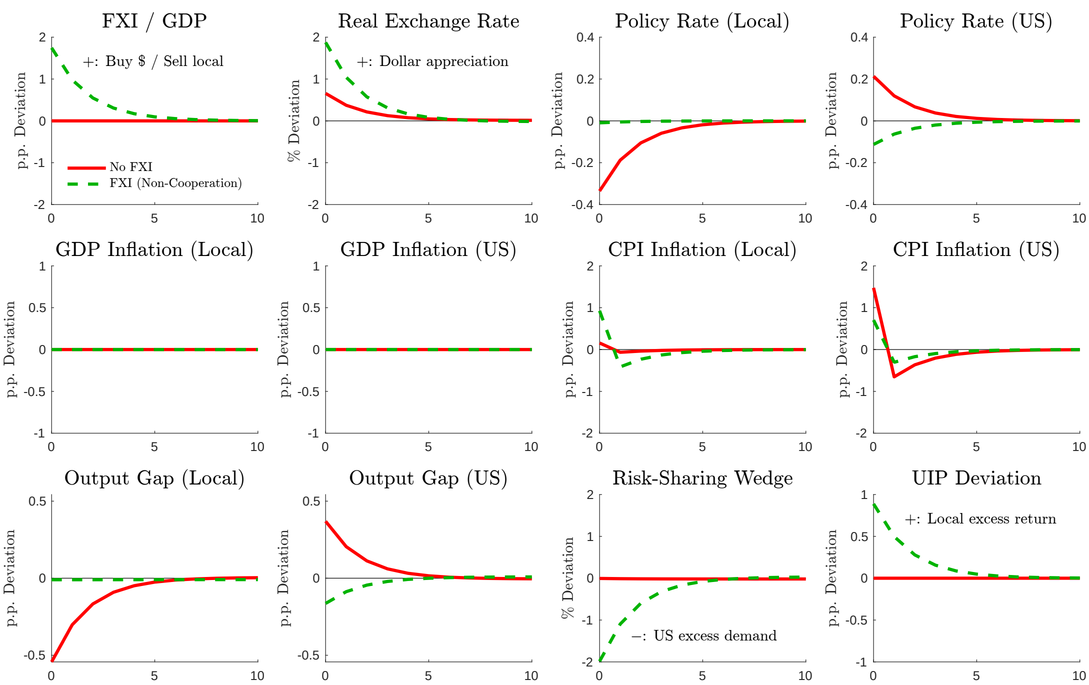


(b) Impulse Responses to a US Cost-Push Inflation



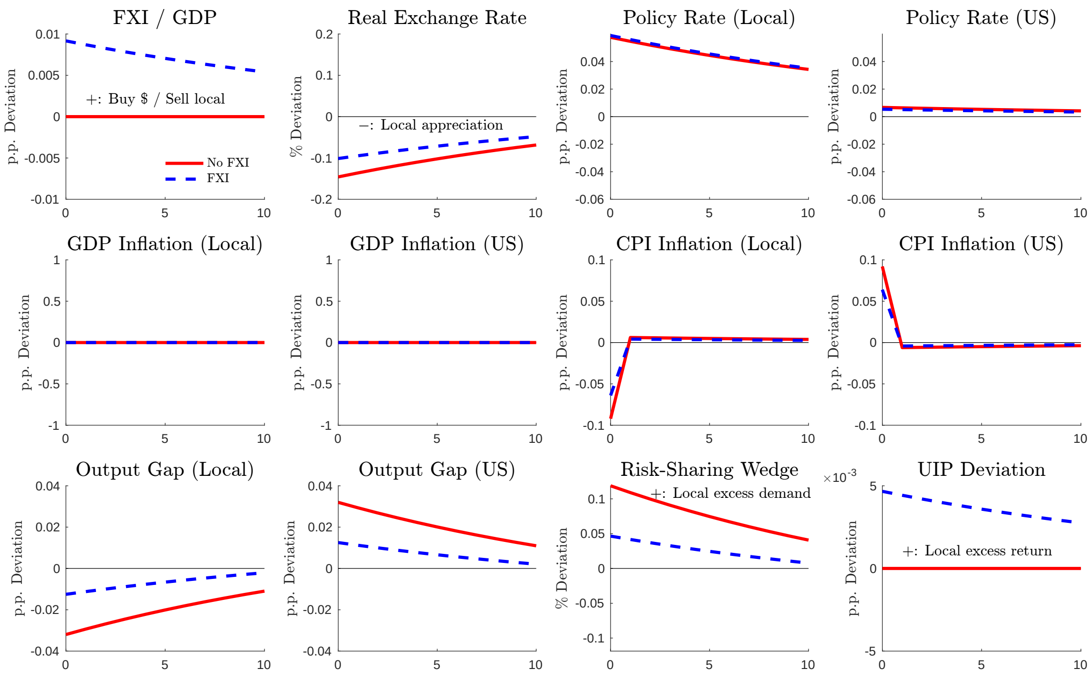
Note: This figure plots impulse responses under DCP and when monetary policy strictly targets inflation. Panel (a) shows the impulse responses to a one-standard-deviation increase in local productivity. Panel (b) shows the impulse responses to a one-standard-deviation increase in US markup. The red and blue lines show the results under no intervention and cooperative intervention cases, respectively.

Figure A4: Impulse Responses to a US Tariff Increase



Note: This figure plots impulse responses to a one-standard-deviation increase in US tariff. The red and green lines show the results under no intervention and non-cooperative intervention cases, respectively.

Figure A5: Impulse Responses to a Positive Local Preference Shock



Note: This figure plots impulse responses to a one-standard-deviation positive shock to the local households' preference. The red and blue lines show the results under no intervention and cooperative intervention cases, respectively.